Transfer Learning Quantile Large-margin Distribution Machine for Wind Power Forecasting under Extreme Events

Junlin Chen[a], Bo Wang[a,*], Zhihang Yu[a], Renju Lin[a], Pei-chun Lin[b], Libo Zhang[c] and Junzo Watada[d]

[a]School of Management and Engineering, Nanjing University, Nanjing 210093, China
[b]Department of Information Engineering and Computer Science, Feng Chia University, Taiwan 40724, China
[c]College of Artificial Intelligence, Southwest University, Chongqing 400715, China
[d]Graduate School of Information, Production and Systems, Waseda University, Kitakyushu 808-0135, Japan

ARTICLE INFO

Keywords:
Wind power forecasting
Large-margin distribution machine
Quantile regression
Transfer learning
Extreme events

ABSTRACT

Accurate wind power forecasting under extreme events is critical for grid stability and renewable energy integration, yet remains a formidable challenge due to severe data scarcity and substantial distribution shifts between normal and extreme-event operating regimes. While kernel-based methods, particularly support vector regression and Large-margin Distribution Machine-based Regression (LDMR), have demonstrated strong performance in various regression tasks due to their theoretical guarantees and robustness on limited data, their application to extreme-event forecasting faces critical limitations: the i.i.d. assumption underlying standard kernel methods is fundamentally violated under cross-domain scenarios, and the symmetric variance minimization in LDMR implicitly encourages symmetric error distributions, which conflicts with the asymmetric nature of quantile-based interval prediction required for uncertainty quantification. To extend kernel-based regression to extreme-event forecasting, we propose Transfer Learning Quantile Large-margin Distribution Machine for Regression (TL-QLDMR), which addresses these challenges through three synergistic innovations: (i) an asymmetric variance regularizer that preserves LDMR's distributional robustness while accommodating quantile-specific error asymmetry, (ii) a supervised transfer learning framework integrating maximum mean discrepancy for feature alignment and weighted pinball loss for label adaptation, enabling effective knowledge transfer from abundant normal-domain data to scarce extreme-event samples, and (iii) a scalable Nyström-SGD solver reducing complexity from $O(N^3)$ to $O(T \cdot N \cdot m)$. Comprehensive experiments on six wind farms demonstrate that TL-QLDMR achieves competitive performance in both deterministic and probabilistic forecasting in the extreme-event domain, with superior reliability-sharpness trade-off and practical scalability, making it suitable for real-time applications where interpretability and theoretical guarantees are prioritized.

1. Introduction

The global transition toward renewable energy has positioned wind power as a cornerstone of sustainable electricity generation. However, the inherent intermittency and volatility of wind resources pose significant challenges to power-system reliability, reserve allocation, and dispatch efficiency (Yuan et al., 2023). Accurate wind power forecasting is therefore critical for reducing balancing costs, supporting storage planning, and enabling secure grid integration (Yan et al., 2025; Wang et al., 2025). While substantial progress has been achieved in short-term forecasting under normal meteorological conditions, prediction performance can degrade significantly during extreme events, particularly those associated with typhoons, cold waves, or heat waves, when wind power fluctuations are most severe and energy-system operations are under the greatest stress (Wang et al., 2024; Jonkers et al., 2024).

Extreme events present a dual challenge for data-driven forecasting models. First, such events are inherently rare, resulting in severe data scarcity: extreme-condition samples typically constitute only a small fraction of

*Corresponding author: Bo Wang
E-mail address: jlchen08@163.com (J. Chen), bowangsme@nju.edu.cn (B. Wang), zhihangyu@smail.nju.edu.cn (Z. Yu), linrenju7@163.com (R. Lin), peiclin@fcu.edu.tw (P. Lin), lbzhang@swu.edu.cn (L. Zhang), junzo.watada.ips@gmail.com (J. Watada)

historical records (Chen & Xu, 2022; Wang et al., 2024). Second, the dynamics under extreme-event conditions differ fundamentally from normal operating regimes, inducing substantial distribution shifts in both feature space (e.g., wind speed, temperature) and label space. In this work, the term extreme events is used in a broad operational sense: it includes, but is not limited to, named severe-weather processes and instead denotes rare, high-risk, and out-of-distribution states induced by extreme meteorological deviations and their associated turbine responses. For instance, during typhoon events, wind speeds may exceed 25 m/s, triggering turbine cut-out operations that produce near-zero power output—a pattern rarely observed in normal-domain training data. This creates a covariate shift in the input space and a label shift in the output space, both of which challenge conventional machine learning models that assume training and testing data are independently and identically distributed (i.i.d.) (Dong et al., 2023; Yin et al., 2021).

Among the diverse landscape of forecasting methodologies, kernel-based approaches—particularly Support Vector Regression (SVR) and its variants—have been extensively applied to wind power forecasting and have demonstrated competitive performance under normal-operating conditions (Wang et al., 2021; Li et al., 2021; Luo et al., 2024; Yuan et al., 2022). The SVR methodology continues to evolve, with recent advances in optimization algorithms, convergence properties, and robustness enhancements. These kernel methods offer several distinctive advantages over black-box deep learning models: strong generalization guarantees rooted in statistical learning theory, effective handling of high-dimensional feature spaces through the kernel trick, convex optimization formulations that guarantee global optima, and robust performance on small-to-medium-sized datasets where deep models tend to overfit. Moreover, their interpretability and theoretical foundations make them particularly attractive for safety-critical applications such as grid dispatch, where model transparency is often required by regulatory standards. Recent studies have further enriched kernel regression through robust SVM formulations (Trafalis & Gilbert, 2006), bounded-loss regression (Fu et al., 2023), and multiple-kernel learning with compound regularization (Jiang et al., 2021), reinforcing the continuing relevance of robustness-oriented kernel learning for forecast tasks involving noisy and heterogeneous data.

Despite these strengths, existing kernel-based wind power forecasting studies predominantly focus on normal-operating conditions with deterministic point prediction objectives. Traditional SVR minimizes the ϵ-insensitive loss to estimate the conditional mean, achieving robust point predictions under Gaussian noise assumptions. More generally, direct quantile-based forecasting has long been recognized as an effective route to interval generation under skewed and heteroscedastic demand patterns (Taylor, 2007). To address the need for uncertainty quantification, Support Vector Quantile Regression (SVQR) extends SVR by replacing the symmetric ϵ-loss with the asymmetric pinball loss, enabling estimation of conditional quantiles for prediction intervals (Ye et al., 2025b, 2023). Recent SVQR variants incorporate twin-hyperplane designs (Ye et al., 2023), robustness-enhancing loss formulations (Hazarika et al., 2023), and nonlinear feature selection (Ye et al., 2025b) to improve scalability and robustness. Despite these methodological advances, the application of kernel-based methods to extreme-event forecasting remains largely unexplored, presenting both challenges and opportunities for methodological extension.

At the methodological level, a fundamental limitation shared by both SVR and SVQR is that they optimize only the minimum margin while ignoring the statistical distribution of margins across the entire dataset. This focus on worst-case samples can lead to overfitting and poor generalization, particularly when training data contains outliers or label noise. To overcome this limitation, the Large-margin Distribution Machine-based Regression (LDMR) paradigm was introduced, which explicitly optimizes both the mean and variance of the margin distribution (Zhang & Zhou, 2020; Gupta & Gupta, 2021). By concentrating prediction errors tightly around zero, LDMR achieves superior generalization compared to classical SVR on a wide range of regression tasks (Gupta & Gupta, 2021; Hazarika & Gupta, 2023). Variants of LDMR have been successfully applied to wind turbine fault detection (Tang et al., 2019), imbalanced classification (Cheng et al., 2016), and structural health monitoring (Yang et al., 2024), demonstrating its robustness

in noisy industrial environments. However, LDMR has not been systematically applied to wind power forecasting, particularly under extreme-event conditions where both distributional robustness and probabilistic interval prediction are required. These characteristics—combined with LDMR's theoretical foundations—make it a promising candidate for extension to forecasting in the extreme-event domain.

Against this backdrop, extending kernel-based methods from the normal-operating domain to the extreme-event domain introduces three critical challenges that require methodological innovation. First, the i.i.d. assumption—that training and testing data are drawn from the same distribution—is a foundational premise of both SVR/SVQR and LDMR. While this assumption holds reasonably well within a single operating regime, it is systematically violated when transferring knowledge from normal (source domain) to extreme-event (target domain) conditions. For example, in our preliminary analysis of six wind farms, the Wasserstein distance between normal and extreme power distributions ranges from 5.96 to 21.06, indicating substantial distributional discrepancy. A model trained solely on abundant normal-domain data will inevitably exhibit performance degradation when deployed under extreme conditions, regardless of how well the margin distribution is optimized within the source domain. Recent wind-power studies have explored transfer learning and domain adaptation to alleviate data scarcity and domain shift in newly built or heterogeneous wind farms (Yin et al., 2021; Wang et al., 2024; Dong et al., 2023); however, their principled integration with kernel-based regression remains an open research question.

Second, there exists a fundamental theoretical tension between LDMR's distributional robustness mechanism and the requirements of quantile regression. The core principle of LDMR—minimizing the variance of prediction residuals—implicitly encourages a symmetric, concentrated error distribution around zero. This is theoretically appropriate and empirically effective for conditional mean estimation, as evidenced by LDMR's superior performance over SVR in numerous regression benchmarks. However, quantile regression at level $\tau \neq 0.5$ requires an inherently asymmetric residual distribution: for example, when estimating the 95th percentile, approximately 95% of errors should be negative and only 5% positive. Naively applying LDMR's variance minimization to quantile regression will force the error distribution toward symmetry, thereby degrading interval calibration and violating the target coverage probability. This theoretical conflict has received limited attention in the literature, as existing LDMR studies focus primarily on mean regression, and existing SVQR studies do not incorporate distributional robustness. Reconciling these two objectives—distributional robustness and quantile-specific asymmetry—requires a principled modification of the LDMR framework.

Third, existing transfer learning approaches for wind power forecasting predominantly adopt unsupervised or weakly supervised adaptation strategies, focusing primarily on aligning feature distributions while underutilizing scarce target-domain labels (Yin et al., 2021; Wang et al., 2024; Dong et al., 2023). This design choice is reasonable when target labels are completely unavailable, but is suboptimal for extreme-event forecasting, where scarce target samples represent high-value information that captures critical domain-specific dynamics such as cut-out behavior and power curtailment. A more effective transfer learning framework should leverage both feature-space alignment and label-space supervision, strategically weighting target samples to guide model adaptation. This supervised transfer learning paradigm has been successfully applied in computer vision and natural language processing, but its integration with kernel-based quantile regression for wind power forecasting remains unexplored.

To address these challenges and extend kernel-based regression to extreme-event forecasting, Transfer Learning Quantile Large-margin Distribution Machine for Regression (TL-QLDMR) is proposed as a principled extension of the classical LDMR paradigm for cross-domain probabilistic wind power prediction. The resulting framework preserves the theoretical strengths of kernel-based methods—convex optimization, generalization guarantees, and robustness on limited data—while introducing three synergistic innovations tailored to the extreme-event setting.

First, a novel asymmetric variance regularization mechanism is introduced to reconcile the tension between distributional robustness and quantile estimation. Unlike standard LDMR, which implicitly encourages symmetric error distributions through variance minimization, the proposed formulation employs asymmetric variance regularization based on algebraic residuals, thereby compressing overall error dispersion while preserving the requisite skewness for accurate quantile prediction. This innovation enables TL-QLDMR to achieve both superior point prediction accuracy and well-calibrated prediction intervals, a capability that neither standard LDMR nor SVQR alone can provide.

Second, a supervised transfer learning framework is developed to integrate two complementary adaptation mechanisms. In the feature space, Maximum Mean Discrepancy (MMD) minimization aligns the source and target distributions in the Reproducing Kernel Hilbert Space (RKHS), reducing covariate shift and ensuring that the learned feature representation is domain-invariant (Tao et al., 2012). In the label space, a weighted pinball loss mechanism assigns substantially higher weights to scarce target-domain samples, treating them as high-value corrective anchors that guide the model toward extreme-event dynamics. This dual-space adaptation strategy enables TL-QLDMR to leverage abundant source data for structural learning while prioritizing target data for domain-specific refinement, thereby addressing the label-neglect problem in existing unsupervised domain adaptation methods. This is particularly important for extreme-event forecasting, where every extreme sample carries valuable information about rare but critical operating conditions such as cut-out events and power curtailment.

Third, because the standard quadratic programming (QP) formulation of kernel-based transfer learning incurs $O(N^3)$ computational complexity through full kernel-matrix computation and inversion, a scalable Nyström-SGD algorithm is developed by combining Nyström approximation—which constructs an explicit low-rank feature mapping using $m \ll N$ landmark points—with stochastic gradient descent (SGD). The resulting complexity is reduced to $O(T \cdot N \cdot m)$, enabling practical deployment on large-scale supervisory control and data acquisition (SCADA) datasets (Chen & Xu, 2022) that accumulate tens of thousands of samples annually.

TL-QLDMR is evaluated on a publicly available multi-site wind farm SCADA dataset encompassing six wind farms with capacities ranging from 36 MW to 200 MW, in comparison with recent kernel-based methods, ensemble approaches, and deep learning baselines. Experimental results demonstrate that TL-QLDMR achieves competitive performance in both deterministic and probabilistic forecasting in the extreme-event domain, with consistent improvements across multiple evaluation metrics. Ablation studies confirm the contribution of each component, and scalability analysis validates the efficiency of the proposed Nyström-SGD solver.

The remainder of this paper is organized as follows. Section 2 reviews the theoretical foundations of SVQR, LDMR, and transfer learning with MMD, establishing the building blocks for our method. Section 3 presents the TL-QLDMR formulation, including the asymmetric variance regularizer, supervised transfer framework, and scalable optimization algorithm. Section 4 describes the dataset, experimental setup, and comprehensive evaluation results across deterministic prediction, probabilistic interval forecasting, ablation studies, and computational efficiency analysis. Finally, Section 5 concludes the paper and discusses future research directions.

2. Related Work

This section briefly reviews the theoretical foundations related to the proposed method, including Support Vector Quantile Regression, Large-margin Distribution Machine-based Regression, and Transfer Learning with Maximum Mean Discrepancy.

2.1. Support Vector Quantile Regression

Traditional SVR relies on the ϵ-insensitive loss function, which assumes a symmetric error distribution (e.g., Gaussian) and estimates the conditional mean of the target variable. However, in scenarios like partial shading or wind curtailment, the noise distribution is often asymmetric and heteroscedastic. To address this, SVQR was introduced to estimate conditional quantiles, providing a comprehensive view of the outcome distribution. Recent SVQR-family developments include twin-hyperplane designs, robustness-enhancing loss formulations, and nonlinear feature-selection variants, which improve scalability and robustness in non-Gaussian environments (Ye et al., 2023; Hazarika et al., 2023; Ye et al., 2025b).

The core of SVQR is the non-symmetric pinball loss function, defined as

$$L_\tau(u) = \begin{cases} \tau u, & u \geq 0 \\ (\tau - 1)u, & u < 0, \end{cases} \tag{1}$$

where $\tau \in (0, 1)$ indicates the target quantile. Unlike SVR, SVQR finds a decision boundary that divides the data such that a proportion τ of the samples lie below the boundary. The optimization problem minimizes the structural risk along with the empirical risk measured by the pinball loss

$$\min_{\boldsymbol{w},b} \frac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_{i=1}^{n} L_\tau(y_i - (\boldsymbol{w}^T\phi(\boldsymbol{x}_i) + b)), \tag{2}$$

this formulation allows SVQR to capture heavy-tailed distributions and outliers more effectively than mean-based regression.

2.2. Large-margin Distribution Machine-based Regression

While SVR and SVQR focus on maximizing the minimum margin, they largely ignore the statistical distribution of the margin over the entire dataset. LDMR was proposed to improve generalization by explicitly optimizing the distribution of the margins. From early large-margin distribution formulations to least-squares and mode-seeking variants, LDMR has evolved into a practical framework for robust learning under noisy conditions (Zhang & Zhou, 2014, 2020; Gupta & Gupta, 2021; Hazarika & Gupta, 2023).

The key idea of LDMR is to not only maximize the margin of the support vectors but also to minimize the variance of the error distribution for all samples. A typical LDMR formulation for regression integrates two additional statistical regularization terms into the objective function

$$\min_{\boldsymbol{w},b} \frac{1}{2}\|\boldsymbol{w}\|^2 + \lambda_1 \mathrm{Var}(\mathrm{Residuals}) + \lambda_2 \mathrm{Mean}(\mathrm{Residuals}) + C\sum_{i=1}^{n} L(y_i, f(\boldsymbol{x}_i)), \tag{3}$$

by constraining the variance, LDMR forces the prediction errors to be clustered tightly around zero, leading to a more compact and robust hypothesis space. However, this variance minimization implicitly encourages symmetric error distributions. Our work extends this concept by introducing an asymmetric variance regularization mechanism suitable for quantile regression, which preserves the distributional robustness of LDMR while accommodating the inherent asymmetry required for accurate quantile estimation.

2.3. Transfer Learning and Maximum Mean Discrepancy

Standard machine learning models operate under the assumption that training (source) and testing (target) data are drawn from the same probability distribution. When this assumption is violated—such as in wind power forecasting across different operating regimes—the model performance degrades significantly. Transfer learning aims to transfer knowledge from a label-rich source domain to a target domain with different but related distributions. MMD-based adaptation has been widely used in transfer scenarios because it provides a non-parametric discrepancy measure in RKHS and integrates naturally with kernelized objectives (Tao et al., 2012; Yan et al., 2022; Guo et al., 2022; Zhai et al., 2023; Se et al., 2023).

A fundamental challenge in transfer learning is measuring and reducing the distribution discrepancy. MMD is a widely used non-parametric metric that measures the distance between two distributions in a RKHS. Given source data $\{\boldsymbol{x}_{S,i}\}$ and target data $\{\boldsymbol{x}_{T,j}\}$, the squared MMD distance is defined as

$$\mathrm{MMD}^2(D_S, D_T) = \left\| \frac{1}{n_S} \sum_{i=1}^{n_S} \phi(\boldsymbol{x}_{S,i}) - \frac{1}{n_T} \sum_{j=1}^{n_T} \phi(\boldsymbol{x}_{T,j}) \right\|_{\mathcal{H}}^2, \tag{4}$$

where $\phi(\cdot)$ is the feature mapping induced by the kernel. By minimizing the MMD distance during training, the model learns a feature representation where the source and target domains are statistically aligned, enabling the classifier or regressor trained on the source domain to generalize well to the target domain. This alignment is particularly crucial for extreme-event forecasting, as it ensures that the learned predictor is robust to the feature-space shift between normal and extreme-event regimes.

3. Proposed Method

This section presents the construction of TL-QLDMR. Necessary notation is introduced first, followed by the final objective function integrating structural risk, distribution variance, domain adaptation, and weighted pinball loss. The constraints required to handle the non-differentiability of the quantile loss are then described.

3.1. Notations

Consider the problem of wind power forecasting under extreme events. The source domain is denoted by $D_S = \{(\boldsymbol{x}_{S,i}, y_{S,i})\}_{i=1}^{n_S}$ and contains abundant normal-operating samples, while the target domain is denoted by $D_T = \{(\boldsymbol{x}_{T,j}, y_{T,j})\}_{j=1}^{n_T}$ and contains scarce extreme-event samples. The input space $\mathcal{X} \subseteq \mathbb{R}^d$ represents feature vectors constructed via sliding window processing, and the output space $\mathcal{Y} \subseteq \mathbb{R}$ represents wind power. The sample sizes satisfy $n_T \ll n_S$.

Due to drastic meteorological changes during extreme events, the joint distributions differ significantly: $P_S(\boldsymbol{X}_S, \boldsymbol{Y}_S) \neq P_T(\boldsymbol{X}_T, \boldsymbol{Y}_T)$. This distribution discrepancy implies that a model trained solely on P_S will fail on P_T. A feature mapping $\phi : \mathcal{X} \to \mathcal{H}$ is used to map the input to a high-dimensional Reproducing Kernel Hilbert Space (RKHS). The prediction model is defined as

$$f(\boldsymbol{x}) = \boldsymbol{w}^T \phi(\boldsymbol{x}) + b, \tag{5}$$

where $\boldsymbol{w}$ is the weight vector and b is the bias. The quantile $\tau \in (0, 1)$ is set to a specific value (e.g., $\tau = 0.95$) for interval prediction (e.g., predicting the upper bound). Unless otherwise noted, $k(\cdot, \cdot)$ denotes the scalar kernel function, $\boldsymbol{K}$ denotes the Gram matrix, $\phi(\cdot)$ denotes the implicit RKHS mapping used in the exact kernel formulation, and $\psi(\cdot)$ denotes the explicit Nyström feature map used only in the scalable primal solver.

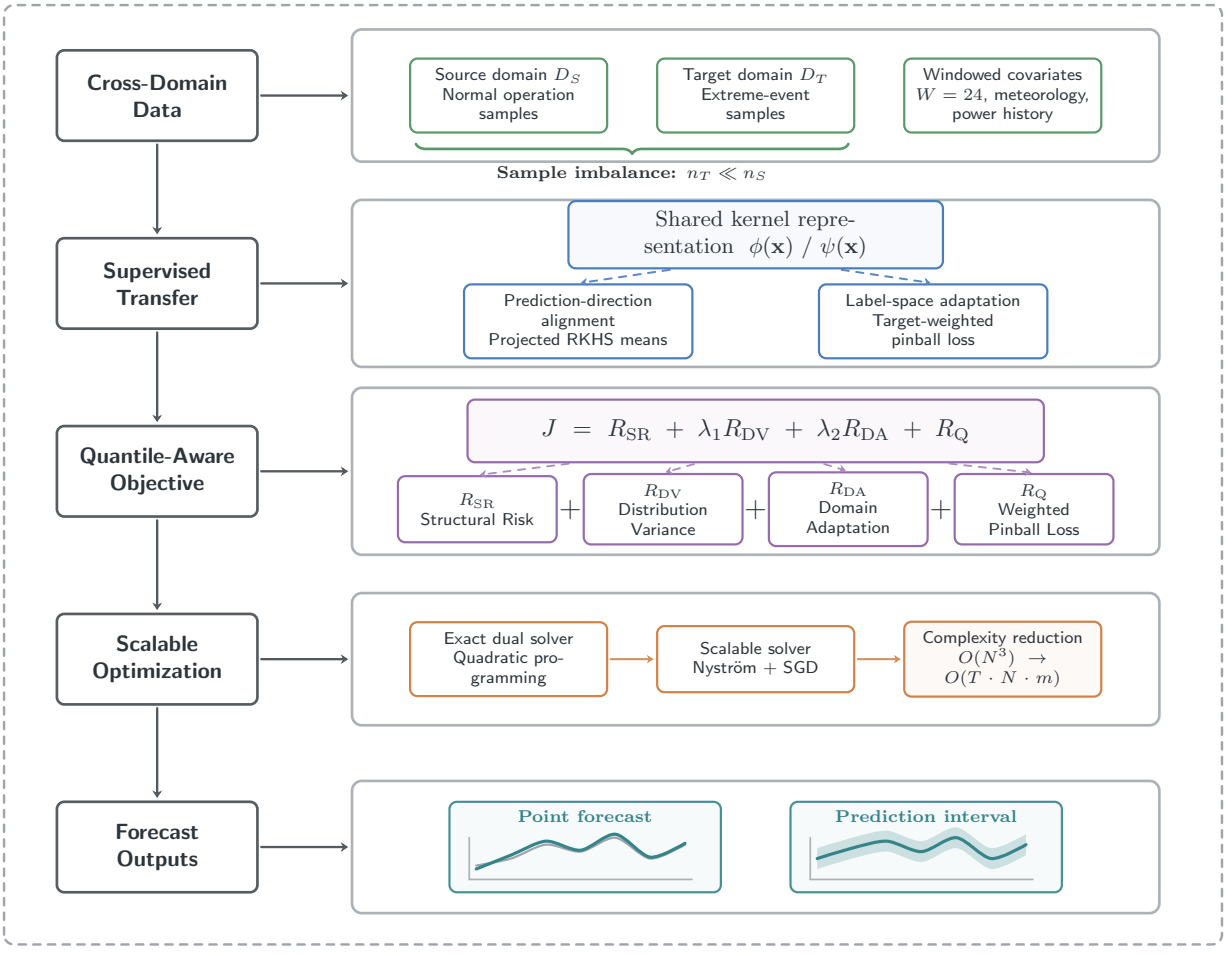


Fig. 1: Overall architecture of the proposed TL-QLDMR model. The left column shows the five-stage pipeline: cross-domain data preparation, supervised transfer learning, quantile-aware objective construction, scalable optimization, and forecast outputs. Each stage is elaborated in the corresponding right panel. In the supervised transfer stage, the shared kernel representation branches into feature-space alignment via MMD in RKHS and label-space adaptation via target-weighted pinball loss (indicated by dashed arrows). The quantile-aware objective decomposes into four additive terms: structural risk R_{SR}, distribution variance R_{DV}, domain adaptation R_{DA}, and weighted pinball loss R_{Q}. The optimization stage provides both an exact dual QP solver and a scalable Nyström-SGD solver with complexity reduction from $O(N^3)$ to $O(T \cdot N \cdot m)$. The final outputs include deterministic point forecasts and probabilistic prediction intervals.

3.2. TL-QLDMR Model Formulation

The overall architecture of the proposed TL-QLDMR model is illustrated in Fig. 1. The left column presents a five-stage pipeline progressing from cross-domain data preparation through supervised transfer, quantile-aware objective construction, scalable optimization, to forecast outputs. In the supervised transfer stage, a shared kernel representation $\phi(\mathbf{x})/\psi(\mathbf{x})$ branches into two complementary adaptation paths: feature-space alignment via MMD in RKHS and label-space adaptation via target-weighted pinball loss. The quantile-aware objective integrates four terms—structural risk, asymmetric distribution variance, MMD-based domain adaptation, and weighted pinball loss—into a unified formulation. The optimization stage supports both an exact dual QP solver and a scalable Nyström-SGD solver, ultimately producing point forecasts and probabilistic prediction intervals.

3.2.1. Objective Function

Our optimization goal is to find the optimal $\boldsymbol{w}, b$ and slack variables $\boldsymbol{\xi}, \boldsymbol{\xi}^*$ to minimize the following objective function J. Let $\bar{u}_S = \frac{1}{n_S}\sum_{i=1}^{n_S}(y_{S,i} - f(\boldsymbol{x}_{S,i}))$ denote the mean residual over the source domain, which is treated as a scalar constant during optimization

$$\begin{aligned}\min_{\boldsymbol{w},b,\boldsymbol{\xi},\boldsymbol{\xi}^*} \quad J = \quad & \underbrace{\frac{1}{2}\|\boldsymbol{w}\|^2}_{\text{(1) Structural Risk}} \\ & + \underbrace{\lambda_1 \sum_{i=1}^{n_S}\left((y_{S,i} - \boldsymbol{w}^T\phi(\boldsymbol{x}_{S,i}) - b) - \bar{u}_S\right)^2}_{\text{(2) Distribution Variance}} \\ & + \underbrace{\lambda_2 \left\| \frac{1}{n_S}\sum_{i=1}^{n_S}\phi(\boldsymbol{x}_{S,i}) - \frac{1}{n_T}\sum_{j=1}^{n_T}\phi(\boldsymbol{x}_{T,j}) \right\|_{\mathcal{H}}^2}_{\text{(3) Domain Adaptation}} \\ & + \underbrace{C_S\sum_{i=1}^{n_S} L_\tau(y_{S,i}, f(\boldsymbol{x}_{S,i})) + C_T\sum_{j=1}^{n_T} L_\tau(y_{T,j}, f(\boldsymbol{x}_{T,j}))}_{\text{(4) Weighted Pinball Loss}},\end{aligned} \tag{6}$$

the physical meaning of each term is as follows:

(1) **Structural Risk**: This term controls the complexity of the model and prevents overfitting, embodying the maximum margin principle of SVM.

(2) **Distribution Variance**: This term is the core of our proposed Quantile LDMR model, incorporating a critical modification to the original LDMR variance.

- **Theoretical Conflict**: The original LDMR minimizes the variance of residuals, which implicitly encourages a symmetric, concentrated error distribution around zero. However, quantile regression at level $\tau \neq 0.5$ requires an asymmetric residual distribution where approximately τ proportion of errors are negative. Applying a symmetric variance penalty would force the residual distribution towards symmetry, degrading the skewness required for accurate quantile estimation.
- **Asymmetric Modification**: To resolve this, we employ an asymmetric variance regularization based on the algebraic residuals. This design allows the model to reduce error dispersion without constraining the distribution's direction, thereby preserving the non-symmetric properties essential for the Pinball Loss and improving interval coverage.

(3) **Domain Adaptation (MMD)**: This term aligns source and target feature distributions in RKHS by minimizing mean-embedding discrepancy. This alignment ensures that the learned predictor is robust to the feature-space shift between normal and extreme-event regimes. MMD is chosen because it is non-parametric, computationally tractable, and directly compatible with kernelized optimization.

(4) **Weighted Pinball Loss**: This term performs label-space adaptation under sample imbalance. Unlike purely unsupervised alignment, the objective explicitly includes target-domain labels:

- **Label-supervised correction**: extreme-event target labels constrain the model in regions where source-domain trends are no longer reliable (e.g., high wind speed under cut-out operation).

- **Imbalance-aware weighting**: by setting $C_T \gg C_S$, prediction errors on scarce target samples receive stronger penalties, preventing optimization from being dominated by abundant source samples.

3.2.2. Constraints

To handle the non-differentiability of the pinball loss, we introduce slack variables ξ_k, ξ_k^* to transform it into linear constraints. These constraints apply uniformly to all samples k in the combined training set $D = D_S \cup D_T$, while domain-specific weighting is handled through C_S and C_T in the objective function

$$\begin{cases} y_k - (\boldsymbol{w}^T \phi(\boldsymbol{x}_k) + b) \le \xi_k, & \forall k \in \{1, \dots, n_S + n_T\} \\ (\boldsymbol{w}^T \phi(\boldsymbol{x}_k) + b) - y_k \le \xi_k^*, & \forall k \in \{1, \dots, n_S + n_T\} \\ \xi_k \ge 0, \quad \xi_k^* \ge 0 \end{cases}, \tag{7}$$

in the objective function, the term $L_\tau(y_k, f(\boldsymbol{x}_k))$ corresponds to $\tau\xi_k + (1-\tau)\xi_k^*$.

3.3. Model Optimization

This section derives the dual problem of TL-QLDMR to enable efficient solution via Quadratic Programming (QP).

3.3.1. Matrix Reformulation of the Primal Problem

To facilitate the derivation, the variance term and the MMD term in the objective function [Eq. (6)] are rewritten using matrix traces or norms. Let $X = [\boldsymbol{X}_S, \boldsymbol{X}_T]$ denote the entire training set with $N = n_S + n_T$ samples. For consistency with standard derivations, a coefficient of $1/2$ is added to the variance and MMD terms. The primal objective can be reformulated as

$$\min_{\boldsymbol{w},b,\xi,\xi^*} \quad \frac{1}{2}\|\boldsymbol{w}\|^2 + \frac{\lambda_1}{2}\mathrm{Var}_S(\boldsymbol{w}, b) + \frac{\lambda_2}{2}\mathrm{MMD}^2(\boldsymbol{w}) + \sum_{k=1}^{N} C_k L_\tau(y_k, f(\boldsymbol{x}_k)), \tag{8}$$

where C_k takes value C_S if sample $k \in D_S$ and C_T if $k \in D_T$.

Two auxiliary matrices are introduced to express the variance and MMD terms in Eq. (8) as quadratic forms of the weights:

(1) **Variance Matrix S**: To represent the source-domain variance $\sum_{i\in D_S}(y_{S,i} - f(\boldsymbol{x}_{S,i}) - \bar{u}_S)^2$, we define a centering matrix $\boldsymbol{H}_S = \boldsymbol{I}_{n_S} - \frac{1}{n_S}\mathbf{1}_{n_S}\mathbf{1}_{n_S}^T$. The variance term can be written as $\|\boldsymbol{H}_S(\boldsymbol{y}_S - \boldsymbol{f}_S)\|^2$. To extend this to the full dataset, we define an $N \times N$ matrix $\mathbf{S}$

$$\mathbf{S} = \begin{bmatrix} \boldsymbol{H}_S & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}. \tag{9}$$

(2) **MMD Matrix M**: The MMD term represents the distance between means. We define the MMD coefficient matrix $\mathbf{M} \in \mathbb{R}^{N\times N}$ as

$$M_{ij} = \begin{cases} \frac{1}{n_S^2} & i, j \in D_S \\ \frac{1}{n_T^2} & i, j \in D_T \\ \frac{-1}{n_S n_T} & \text{otherwise} \end{cases}. \tag{10}$$

3.3.2. Representer Theorem and Kernelization

According to the representer theorem (Schölkopf et al. (2001)), the optimal weight vector $\boldsymbol{w}$ lies in the span of the feature mappings of the training samples. Note that the representer theorem applies here because all regularization terms (structural risk, variance, and MMD) can be expressed as functions of inner products in RKHS, preserving the kernel structure. Therefore

$$\boldsymbol{w} = \sum_{i=1}^{N} \beta_i \phi(\boldsymbol{x}_i) = \boldsymbol{\Phi}\boldsymbol{\beta}, \tag{11}$$

where $\boldsymbol{\Phi} = [\phi(\boldsymbol{x}_1), \dots, \phi(\boldsymbol{x}_N)]$ is the feature matrix and $\boldsymbol{\beta} = [\beta_1, \dots, \beta_N]^T$ are the coefficients. Substituting Eq. (11) into the prediction function, we get $f(\boldsymbol{X}) = \boldsymbol{\beta}^T \boldsymbol{K} + b\mathbf{1}^T$, where $\boldsymbol{K} = \boldsymbol{\Phi}^T\boldsymbol{\Phi}$ is the $N \times N$ kernel matrix. Consequently, using Eq. (9) and Eq. (10), the structural risk, variance, and MMD terms can be expressed in terms of $\boldsymbol{\beta}$:

- Structural Risk: $\|\boldsymbol{w}\|^2 = \boldsymbol{\beta}^T \boldsymbol{K}\boldsymbol{\beta}$.
- Distribution Variance: The quadratic part with respect to $\boldsymbol{\beta}$ is $\boldsymbol{\beta}^T \boldsymbol{K}\boldsymbol{S}\boldsymbol{K}\boldsymbol{\beta}$. Note that the bias b is eliminated by the centering matrix $\boldsymbol{H}_S$.
- Domain Adaptation (MMD): $\text{MMD}^2(\boldsymbol{w}) = \boldsymbol{\beta}^T \boldsymbol{K}\boldsymbol{M}\boldsymbol{K}\boldsymbol{\beta}$.

Combining these, we define the total coefficient matrix $\boldsymbol{\Omega}$

$$\boldsymbol{\Omega} = \boldsymbol{K} + \lambda_1 \boldsymbol{K}\boldsymbol{S}\boldsymbol{K} + \lambda_2 \boldsymbol{K}\boldsymbol{M}\boldsymbol{K}. \tag{12}$$

Remark 1. *When $\boldsymbol{K}$ is positive definite (which holds for common kernels such as RBF on distinct samples) and $\lambda_1, \lambda_2 > 0$, the matrix $\boldsymbol{\Omega}$ is positive definite. This ensures that $\boldsymbol{\Omega}$ is invertible and the optimization problem is well-posed. The positive definiteness of $\boldsymbol{\Omega}$ also guarantees the strong convexity of the objective function, which is essential for both the dual QP formulation and the convergence analysis of the scalable SGD algorithm.*

By substituting the kernelized terms and Eq. (12) into Eq. (8), the kernelized primal problem becomes (ignoring constant terms w.r.t b in variance)

$$\min_{\boldsymbol{\beta}, b, \boldsymbol{\xi}, \boldsymbol{\xi}^*} \quad \frac{1}{2}\boldsymbol{\beta}^T \boldsymbol{\Omega}\boldsymbol{\beta} - (\lambda_1 \boldsymbol{K}\boldsymbol{S}\boldsymbol{Y})^T \boldsymbol{\beta} + \sum_{k=1}^{N} C_k(\tau\xi_k + (1-\tau)\xi_k^*), \tag{13}$$

subject to the constraints from Eq. (7)

$$y_k - (\boldsymbol{K}_k\boldsymbol{\beta} + b) \le \xi_k, \quad (\boldsymbol{K}_k\boldsymbol{\beta} + b) - y_k \le \xi_k^*, \quad \xi_k, \xi_k^* \ge 0. \tag{14}$$

3.3.3. Lagrangian and Dual Problem

We introduce Lagrange multipliers $\alpha_k, \alpha_k^*, \mu_k, \mu_k^* \ge 0$ for Eq. (14). The Lagrangian function $\mathcal{L}$ is constructed as

$$\begin{aligned} \mathcal{L} =& \frac{1}{2}\boldsymbol{\beta}^T \boldsymbol{\Omega}\boldsymbol{\beta} - (\lambda_1 \boldsymbol{K}\boldsymbol{S}\boldsymbol{Y})^T \boldsymbol{\beta} + \sum_{k=1}^{N} C_k(\tau\xi_k + (1-\tau)\xi_k^*) \\ & - \sum_{k=1}^{N} \alpha_k(\xi_k - y_k + \boldsymbol{K}_k\boldsymbol{\beta} + b) - \sum_{k=1}^{N} \alpha_k^*(\xi_k^* + y_k - \boldsymbol{K}_k\boldsymbol{\beta} - b) - \sum(\mu_k\xi_k + \mu_k^*\xi_k^*), \end{aligned} \tag{15}$$

by setting the partial derivatives of Eq. (15) with respect to the primal variables $(b, \boldsymbol{\xi}, \boldsymbol{\xi}^*, \boldsymbol{\beta})$ to zero, along with the complementary slackness and primal feasibility conditions, we obtain the KKT conditions

$$\frac{\partial \mathcal{L}}{\partial b} = 0 \implies \sum_{k=1}^{N}(\alpha_k - \alpha_k^*) = 0 \tag{16}$$

$$\frac{\partial \mathcal{L}}{\partial \xi_k} = 0, \frac{\partial \mathcal{L}}{\partial \xi_k^*} = 0 \implies \alpha_k \in [0, C_k\tau], \quad \alpha_k^* \in [0, C_k(1-\tau)] \tag{17}$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\beta}} = 0 \implies \boldsymbol{\Omega\beta} - \lambda_1 \boldsymbol{KSY} - \boldsymbol{K}(\boldsymbol{\alpha} - \boldsymbol{\alpha}^*) = 0, \tag{18}$$

let $\boldsymbol{\gamma} = \boldsymbol{\alpha} - \boldsymbol{\alpha}^*$. From Eq. (18), the optimal $\boldsymbol{\beta}$ is derived as

$$\boldsymbol{\beta} = \boldsymbol{\Omega}^{-1}(\boldsymbol{K\gamma} + \lambda_1 \boldsymbol{KSY}). \tag{19}$$

Substituting Eq. (19) back into Eq. (15), and simplifying using Eq. (16) and Eq. (17), we arrive at the dual Problem, which is a standard Quadratic Programming (QP) problem

$$\begin{aligned} \min_{\boldsymbol{\alpha},\boldsymbol{\alpha}^*} \quad & \frac{1}{2}(\boldsymbol{\alpha} - \boldsymbol{\alpha}^*)^T \mathbf{Q}_{dual}(\boldsymbol{\alpha} - \boldsymbol{\alpha}^*) + \mathbf{p}_{dual}^T(\boldsymbol{\alpha} - \boldsymbol{\alpha}^*) \\ \text{s.t.} \quad & \sum_{k=1}^{N}(\alpha_k - \alpha_k^*) = 0 \\ & 0 \le \alpha_k \le C_k\tau, \quad \forall k \\ & 0 \le \alpha_k^* \le C_k(1-\tau), \quad \forall k \end{aligned}, \tag{20}$$

where the dual matrix $\mathbf{Q}_{dual}$ and vector $\mathbf{p}_{dual}$ are defined as

$$\mathbf{Q}_{dual} = \boldsymbol{K\Omega}^{-1}\boldsymbol{K}, \quad \mathbf{p}_{dual} = \lambda_1 \boldsymbol{K\Omega}^{-1}\boldsymbol{KSY} - \boldsymbol{Y}. \tag{21}$$

Once the optimal $\boldsymbol{\alpha}, \boldsymbol{\alpha}^*$ are found using a QP solver for Eq. (20), we can recover $\boldsymbol{\beta}$ using Eq. (19) and construct the final decision function $f(\boldsymbol{x})$.

3.4. The Algorithm and Complexity Analysis

The complete training process of the proposed TL-QLDMR model is summarized in Algorithm 1. The algorithm mainly consists of matrix construction and solving a QP problem.

The computational complexity of TL-QLDMR primarily stems from three parts: matrix multiplication, matrix inversion, and the QP solver.

(1) **Matrix Operations**: The construction of $\boldsymbol{K}$, $\mathbf{S}$, and $\mathbf{M}$ involves $O(N^2 d)$ or $O(N^2)$ operations. The calculation of $\boldsymbol{\Omega}$ involves matrix multiplications of size $N \times N$, leading to a complexity of $O(N^3)$ in the worst case for dense kernel matrices.

(2) **Matrix Inversion**: Inverting the $N \times N$ matrix $\boldsymbol{\Omega}$ requires $O(N^3)$ complexity in the worst case.

(3) **QP Solver**: Solving the standard Quadratic Programming problem for the dual variables typically has a complexity ranging from $O(N^{2.3})$ to $O(N^3)$ depending on the solver implementation (e.g., Interior Point Method).

Algorithm 1: TL-QLDMR Training Algorithm

Input : Source Data $D_S = \{(\boldsymbol{x}_{S,i}, y_{S,i})\}_{i=1}^{n_S}$, Target Data $D_T = \{(\boldsymbol{x}_{T,j}, y_{T,j})\}_{j=1}^{n_T}$, Quantile τ, Kernel parameters, Regularization parameters λ_1, λ_2, Pinball Loss weights C_S, C_T.

Output: Optimal dual variables $\boldsymbol{\alpha}, \boldsymbol{\alpha}^*$ and bias b.

`// Step 1: Data Preparation`

Combine Source and Target data: $\boldsymbol{X} = [\boldsymbol{X}_S, \boldsymbol{X}_T]$, $\boldsymbol{Y} = [\boldsymbol{Y}_S; \boldsymbol{Y}_T]$, $N = n_S + n_T$;

Set weights vector $\boldsymbol{C} \in \mathbb{R}^N$ where $C_k = C_S$ for $k \le n_S$ and C_T otherwise;

`// Step 2: Matrix Construction`

Compute Kernel Matrix $\boldsymbol{K} \in \mathbb{R}^{N \times N}$ using the specified kernel function;

Construct Variance Matrix $\mathbf{S}$ using Eq. (9);

Construct MMD Matrix $\mathbf{M}$ using Eq. (10);

`// Step 3: Dual Problem Formulation`

Compute coefficient matrix $\boldsymbol{\Omega} = \boldsymbol{K} + \lambda_1 \boldsymbol{K}\mathbf{S}\boldsymbol{K} + \lambda_2 \boldsymbol{K}\mathbf{M}\boldsymbol{K}$ using Eq. (12);

Invert $\boldsymbol{\Omega}$ to fit the dual form;

Compute Dual Matrix $\mathbf{Q}_{dual} = \boldsymbol{K}\boldsymbol{\Omega}^{-1}\boldsymbol{K}$ using Eq. (21);

Compute Dual Vector $\mathbf{p}_{dual} = \lambda_1 \boldsymbol{K}\boldsymbol{\Omega}^{-1}\boldsymbol{K}\mathbf{S}\boldsymbol{Y} - \boldsymbol{Y}$ using Eq. (21);

`// Step 4: Solve QP`

Solve the QP problem in Eq. (20) providing $\mathbf{Q}_{dual}$, $\mathbf{p}_{dual}$, and constraints $0 \le \alpha_k \le C_k\tau$, $0 \le \alpha_k^* \le C_k(1-\tau)$;

Return Optimal $\boldsymbol{\alpha}, \boldsymbol{\alpha}^*$;

`// Step 5: Recover Primal Variables`

Compute $\boldsymbol{\gamma} = \boldsymbol{\alpha} - \boldsymbol{\alpha}^*$;

Recover $\boldsymbol{\beta}$ using Eq. (19);

Compute bias b using complementary slackness conditions (averaged over Support Vectors);

Therefore, the overall time complexity of the proposed algorithm is approximately $O(N^3)$ in the worst case. Given that $N = n_S + n_T$ and $n_T \ll n_S$, the scale is dominated by the source-domain sample size. While this complexity is acceptable for moderate-sized datasets, it becomes prohibitive for large-scale wind power forecasting scenarios involving tens of thousands of samples. To overcome this computational bottleneck and enable practical deployment in real-world wind farm applications, we develop a scalable optimization algorithm in the following subsection.

3.5. Scalable Optimization via Nyström Approximation

To address the computational bottleneck of the standard QP-based solver, we propose a scalable Nyström-SGD algorithm. This approach reduces the time complexity from $O(N^3)$ to $O(T \cdot N \cdot m)$, where T denotes the number of iterations and $m \ll N$ is the number of landmark points.

3.5.1. Nyström Feature Mapping

The core idea is to approximate the implicit infinite-dimensional feature mapping $\phi(\boldsymbol{x})$ with an explicit finite-dimensional representation. Given the training data, we first select m landmark points $Z = \{z_1, \dots, z_m\}$ from the source domain D_S using K-Means clustering to capture the underlying data manifold, following the rationale of clustered Nyström landmark selection (Zhang & Kwok, 2010).

Let $\boldsymbol{W} \in \mathbb{R}^{m \times m}$ denote the kernel matrix of landmarks, where $W_{ij} = k(z_i, z_j)$. The explicit feature mapping $\psi : \mathcal{X} \to \mathbb{R}^m$ is then constructed as

$$\psi(\boldsymbol{x}) = \boldsymbol{W}^{-1/2}[k(\boldsymbol{x}, z_1), \dots, k(\boldsymbol{x}, z_m)]^T, \tag{22}$$

where $\boldsymbol{W}^{-1/2}$ is computed via eigendecomposition. This mapping satisfies the approximation property $k(\boldsymbol{x}_i, \boldsymbol{x}_j) \approx \psi(\boldsymbol{x}_i)^T \psi(\boldsymbol{x}_j)$, enabling us to work directly in the primal space.

3.5.2. Primal Formulation with Explicit Features

With the explicit feature mapping, the prediction function becomes $f(\boldsymbol{x}) = \boldsymbol{w}^T\psi(\boldsymbol{x}) + b$, where $\boldsymbol{w} \in \mathbb{R}^m$. Let $\boldsymbol{\Psi}_S \in \mathbb{R}^{n_S \times m}$ and $\boldsymbol{\Psi}_T \in \mathbb{R}^{n_T \times m}$ denote the mapped feature matrices for source and target domains, respectively. The regularization matrix $\boldsymbol{R} \in \mathbb{R}^{m \times m}$ incorporating structural risk, variance, and MMD terms is formulated as

$$\boldsymbol{R} = \boldsymbol{I} + \lambda_1 \underbrace{\boldsymbol{\Psi}_S^T \boldsymbol{H}_S \boldsymbol{\Psi}_S}_{\text{Variance}} + \lambda_2 \underbrace{(\bar{\boldsymbol{\psi}}_S - \bar{\boldsymbol{\psi}}_T)(\bar{\boldsymbol{\psi}}_S - \bar{\boldsymbol{\psi}}_T)^T}_{\text{MMD}}, \tag{23}$$

where $\boldsymbol{H}_S = \boldsymbol{I}_{n_S} - \frac{1}{n_S}\mathbf{1}_{n_S}\mathbf{1}_{n_S}^T$ is the centering matrix, and $\bar{\boldsymbol{\psi}}_S$, $\bar{\boldsymbol{\psi}}_T$ are the mean feature vectors of source and target domains.

3.5.3. Stochastic Gradient Descent Optimization

Instead of solving the dual QP problem, we directly optimize the primal objective using the Nyström-SGD procedure. For each sample k in a mini-batch $\mathcal{B}$, the sub-gradient of the Pinball Loss with respect to the prediction is

$$g_k = \frac{\partial L_\tau}{\partial \hat{y}_k} = \begin{cases} \tau - 1, & \text{if } y_k < \hat{y}_k \\ \tau, & \text{if } y_k \geq \hat{y}_k. \end{cases} \tag{24}$$

The weighted sub-gradient is computed as $\tilde{g}_k = C_k \cdot g_k$, where $C_k \in \{C_S, C_T\}$ depends on the domain membership. The parameter gradients are then

$$\nabla_{\boldsymbol{w}} J = \frac{1}{|\mathcal{B}|} \sum_{k \in \mathcal{B}} \tilde{g}_k \psi(\boldsymbol{x}_k) + \boldsymbol{R}\boldsymbol{w}, \quad \nabla_b J = \frac{1}{|\mathcal{B}|} \sum_{k \in \mathcal{B}} \tilde{g}_k. \tag{25}$$

The complete training procedure is summarized in Algorithm 2.

3.5.4. Complexity Analysis of the Scalable Algorithm

The proposed scalable algorithm achieves significant computational savings:

(1) **Landmark Selection**: K-Means clustering requires $O(N \cdot m \cdot I_{km})$ operations, where I_{km} is the number of K-Means iterations.

(2) **Feature Mapping**: Computing $\boldsymbol{W}^{-1/2}$ costs $O(m^3)$, and mapping all N samples requires $O(N \cdot m \cdot d)$.

(3) **Regularization Matrix**: Constructing $\boldsymbol{R}$ involves $O(n_S \cdot m^2)$ for the variance term and $O(m^2)$ for the MMD term.

(4) **SGD Iterations**: Each iteration processes $O(N)$ samples with $O(m)$ operations per sample, yielding $O(T \cdot N \cdot m)$ total complexity.

The overall time complexity is $O(T \cdot N \cdot m + m^3)$. Since $m \ll N$ (typically $m = O(\sqrt{N})$), this represents a substantial reduction from the $O(N^3)$ complexity of the standard QP solver. Table 1 summarizes the comparison between the two algorithms.

3.5.5. Theoretical Justification

The convergence of the proposed Nyström-SGD algorithm is guaranteed under standard assumptions. Since the Pinball Loss is convex and Lipschitz continuous, and the regularization term $\frac{1}{2}\boldsymbol{w}^T\boldsymbol{R}\boldsymbol{w}$ is strongly convex (as $\boldsymbol{R}$ inherits positive definiteness from $\boldsymbol{\Omega}$ when $\lambda_1, \lambda_2 > 0$), the overall objective function is strongly convex. According to the

Algorithm 2: TL-QLDMR via Nyström-SGD

Input : Source data D_S, Target data D_T, Number of landmarks m, Quantile τ, Hyperparameters $\lambda_1, \lambda_2, C_S, C_T$, Learning rate η, Batch size B, Max iterations T.

Output: Optimal weights $\boldsymbol{w} \in \mathbb{R}^m$ and bias $b \in \mathbb{R}$.

`// Step 1: Nyström Feature Mapping`

Select m landmarks $Z = \{z_1, \ldots, z_m\}$ from D_S via K-Means;

Compute landmark kernel matrix $\boldsymbol{W} \in \mathbb{R}^{m \times m}$;

Compute $\boldsymbol{W}^{-1/2}$ via eigendecomposition;

Construct explicit features $\psi(\boldsymbol{x})$ for all samples using Eq. (22);

`// Step 2: Pre-compute Regularization Matrix`

Compute feature matrices $\boldsymbol{\Psi}_S, \boldsymbol{\Psi}_T$ and mean vectors $\bar{\boldsymbol{\psi}}_S, \bar{\boldsymbol{\psi}}_T$;

Construct regularization matrix $\boldsymbol{R}$ using Eq. (23);

`// Step 3: Stochastic Optimization`

Initialize $\boldsymbol{w} \leftarrow \boldsymbol{0}$, $b \leftarrow 0$;

for $t = 1$ **to** T **do**

  Shuffle combined dataset $D = D_S \cup D_T$;

  for *each mini-batch* $\mathcal{B} \subset D$ **do**

    Compute predictions: $\hat{y}_k = \boldsymbol{w}^T \psi(\boldsymbol{x}_k) + b, \forall k \in \mathcal{B}$;

    Compute sub-gradients g_k using Eq. (24);

    Apply sample weights: $\tilde{g}_k \leftarrow C_k \cdot g_k$;

    Compute gradients $\nabla_{\boldsymbol{w}} J$, $\nabla_b J$ using Eq. (25);

    Update $\boldsymbol{w}$, b using the stochastic gradient step with learning rate η;

  end

end

return $\boldsymbol{w}$, b;

Table 1
Complexity comparison between standard QP and scalable algorithm

Component	Standard QP	Scalable (Ours)
Kernel/Feature Computation	$O(N^2 d)$	$O(Nmd)$
Matrix Inversion	$O(N^3)$	$O(m^3)$
Optimization	$O(N^3)$	$O(TNm)$
Total	$O(N^3)$	$O(TNm + m^3)$

convergence theory of SGD for strongly convex functions (Bottou et al., 2018), the algorithm converges to the global optimum at a rate of $O(1/T)$.

Furthermore, the Nyström approximation error is bounded. Let $\tilde{k}(\boldsymbol{x}_i, \boldsymbol{x}_j) = \psi(\boldsymbol{x}_i)^T \psi(\boldsymbol{x}_j)$ denote the approximated kernel. Standard analyses of the Nyström method show that the approximation error decreases with the number of landmarks under suitable spectral assumptions and sampling schemes (Kumar et al., 2012; Zhang & Kwok, 2010). A representative order characterization is

$$\mathbb{E}\left[\|\boldsymbol{K} - \tilde{\boldsymbol{K}}\|_F\right] \leq O\left(\frac{N}{\sqrt{m}}\right), \tag{26}$$

this bound ensures that with a sufficient number of landmarks, the approximation quality is well-controlled, and the solution obtained by the scalable algorithm closely approximates that of the exact QP solver.

Table 2
Summary of wind farm sites used in the experiments.

Farm ID	Site Name	Nominal Capacity (MW)	Total Samples
1	Wind farm site 1	99	~70,000
2	Wind farm site 2	200	~70,000
3	Wind farm site 3	99	~70,000
4	Wind farm site 4	66	~70,000
5	Wind farm site 5	36	~70,000
6	Wind farm site 6	96	~70,000

4. Experiments

In this section, we conduct a comprehensive experimental evaluation of the proposed TL-QLDMR model for wind power forecasting in the extreme-event domain. The experiments are organized into five progressive stages: (1) extreme-event sample screening and domain shift quantification across multiple wind farms (Section 4.5); (2) deterministic point prediction comparison on the wind farm exhibiting the most severe distribution shift (Section 4.6); (3) probabilistic interval prediction evaluation under a 95% confidence level (Section 4.7); (4) ablation study to validate the contribution of each core component (Section 4.8); and (5) scalable optimization validation to assess the computational efficiency of the proposed Nyström-SGD solver (Section 4.9).

4.1. Dataset Description

The experiments are conducted on a publicly available wind farm SCADA dataset published by the State Grid Renewable Energy Generation Dataset (Chen & Xu, 2022). The dataset contains supervisory control and data acquisition records from six onshore wind farms with nominal capacities ranging from 36 MW to 200 MW. Each record is sampled at 15-minute intervals and includes the following meteorological and operational variables:

- Wind speed at multiple heights: hub height, 50 m, 30 m, and 10 m (m/s);
- Wind direction at hub height (degrees);
- Air temperature (°C);
- Atmospheric pressure (hPa);
- Relative humidity (%);
- Active power output (MW).

Table 2 summarizes the basic characteristics of the six wind farms. Each farm contains approximately 70,000 time-stamped records spanning multiple years of continuous operation.

4.2. Feature Engineering and Data Preprocessing

The feature set comprises multi-height wind speeds, air temperature, atmospheric pressure, relative humidity, and cyclically encoded wind direction. Specifically, the raw wind direction angle θ is transformed into two orthogonal components $\sin(\theta)$ and $\cos(\theta)$ to eliminate the discontinuity at the $0°/360°$ boundary, which is a well-established practice in meteorological feature engineering (Wang et al., 2021).

A sliding window of length $W = 24$ (corresponding to a 6-hour lookback horizon at 15-minute resolution) is applied to construct supervised learning samples. This choice provides sufficient short-term temporal context to capture

local persistence and ramping behavior, while avoiding the excessive input dimensionality and redundancy that longer windows may introduce in kernel-based models. For each window, the feature matrix $\boldsymbol{X}_t \in \mathbb{R}^{W\times d}$ is flattened into a vector $\boldsymbol{x}_t \in \mathbb{R}^{Wd}$, and the target variable y_t is the power output at the next time step. All features and the target variable are standardized to zero mean and unit variance using statistics computed from the training set. Missing values are imputed with the per-feature mean to ensure numerical stability for kernel-based models.

4.3. Extreme-Event Domain Definition

A critical prerequisite for evaluating transfer learning under extreme conditions is the rigorous definition of what constitutes an "extreme event" in wind power forecasting (Sideratos & Hatziargyriou, 2012; Liu et al., 2024). In this study, extreme events do not denote a single named meteorological subtype, such as typhoon, icing, or a specific ramping pattern. Instead, they refer to a unified target domain composed of rare, high-risk, high-volatility, and out-of-distribution operating states induced by extreme meteorological deviations and their associated turbine responses relative to the normal-operating domain, consistent with transfer-learning settings in which the source and target domains differ through their marginal distributions and operational regimes (Pan & Yang, 2010; Ben-David et al., 2010). To construct this target domain, a physics-statistics hybrid screening strategy is adopted: physical thresholding captures deterministic high-wind shutdown risk, whereas percentile-based rules identify statistically rare meteorological deviations at each wind farm, following the broader logic of mixed rule-and-statistics criteria used in recent extreme-event energy data studies (Liu et al., 2023). A sample is classified as belonging to the target domain if it satisfies any of the following criteria:

(1) **Statistical wind speed extreme**: Hub-height wind speed exceeds the 98th percentile of the farm's historical distribution;
(2) **Temperature extreme**: Air temperature falls below the 2nd percentile or exceeds the 98th percentile;
(3) **Cut-out wind speed**: Hub-height wind speed exceeds 25 m/s, the typical cut-out threshold for modern wind turbines.

The 25 m/s threshold is treated here as a nominal cut-out speed. Although the exact cut-out point may vary slightly across turbine types, values around 25 m/s are widely used in wind turbine power-curve characterization as the shutdown region associated with safety protection (Rabanal et al., 2019). This threshold is therefore sufficient to capture the core methodological feature of high-wind-induced nonlinear shutdown risk. Notably, the power ramp criterion (rapid power fluctuations exceeding 5% of nominal capacity per 15-minute interval) is deliberately excluded from the extreme definition. This design choice avoids conflating control-induced power curtailment or dispatch adjustments with genuinely weather-driven anomalies.

No further subdivision of extreme-event samples into multiple subclasses is attempted. This choice is deliberate because extreme events are inherently scarce in realistic forecasting settings; splitting them into fine-grained categories would substantially weaken the statistical support available for transfer learning and would turn a small-sample problem into a near-few-shot problem, a difficulty also reported in recent extreme-condition wind power studies (Liang et al., 2025). From the perspective of domain adaptation, these heterogeneous extreme states share the same modeling characteristic: they are all out-of-distribution samples exhibiting elevated volatility, non-stationarity, and clear deviation from normal operating patterns (Pan & Yang, 2010; Ben-David et al., 2010). The objective of TL-QLDMR is therefore to learn a unified disturbance-robust adaptation mechanism rather than a separate submodel for each rare event type. The union of the above three rules thus forms a single target domain that is physically interpretable, statistically sufficient for model training, and consistent with the data-scarce nature of extreme-event forecasting. All remaining samples constitute the normal-operating source domain.

Table 3
Domain shift metrics between normal-operating and extreme-event domains across six wind farms. Farms are sorted by the composite Wasserstein distance in descending order.

Wind Farm	Extreme-Event Samples	W_1(Wind)	W_1(Power)	KS(Wind)	KS(Power)
Farm 6 (96 MW)	4,144	4.060	21.057	0.389	0.344
Farm 3 (99 MW)	4,144	3.389	18.915	0.338	0.265
Farm 2 (200 MW)	4,211	3.467	16.166	0.333	0.142
Farm 5 (36 MW)	3,039	4.195	9.851	0.460	0.363
Farm 4 (66 MW)	4,046	4.235	7.759	0.347	0.201
Farm 1 (99 MW)	4,184	3.140	5.964	0.335	0.116

4.4. Experimental Setup

For deterministic point prediction, we employ Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2) as evaluation metrics. RMSE and MAE quantify the magnitude of prediction errors, while R^2 measures the proportion of variance explained by the model. For probabilistic interval prediction, we adopt Prediction Interval Coverage Probability (PICP), Mean Prediction Interval Width (MPIW), Prediction Interval Normalized Average Width (PINAW), Coverage Width-based Criterion (CWC), and Winkler Score. PICP measures reliability (whether the interval achieves the target coverage), while MPIW and PINAW measure sharpness (how tight the interval is). CWC is a composite metric that penalizes PINAW when PICP falls below the nominal coverage $1-\alpha$, and the Winkler Score combines interval width with a penalty proportional to the magnitude of coverage violations. An ideal model achieves PICP $\geq 1-\alpha$ with the smallest possible PINAW.

To quantify the distribution discrepancy between the normal-operating and extreme-event domains, we employ two complementary metrics computed on both wind speed and power dimensions: the Wasserstein-1 distance (W_1), which measures the minimum transportation cost to transform one distribution into the other, and the Kolmogorov–Smirnov statistic (KS), which captures the supremum of the absolute difference between the empirical cumulative distribution functions of the two domains.

The hyperparameters of TL-QLDMR are selected by validation-based tuning within each experiment. The searched ranges are logarithmic grids for λ_1 and λ_2 from 10^{-6} to 10^{6}, $C_S \in \{0.05, 0.1, 0.2\}$, $C_T \in \{50, 300, 1000\}$, and kernel bandwidth $\gamma \in \{2^{-6}, \ldots, 2^{6}\}$. The Nyström-SGD solver is tuned over landmark size $m \in [800, 1500]$, learning rate in $\{0.005, 0.01\}$, and training epochs in $[48, 80]$, with mini-batch size fixed at 256. All experiments are implemented in Python and executed on a GPU server equipped with eight NVIDIA Tesla V100-SXM2-32GB GPUs (32 GB memory per GPU), using NVIDIA driver version 550.54.15 and CUDA 12.4. All random splits use a fixed seed for reproducibility.

4.5. Experiment 1: Extreme-Event Sample Screening and Domain Shift Quantification

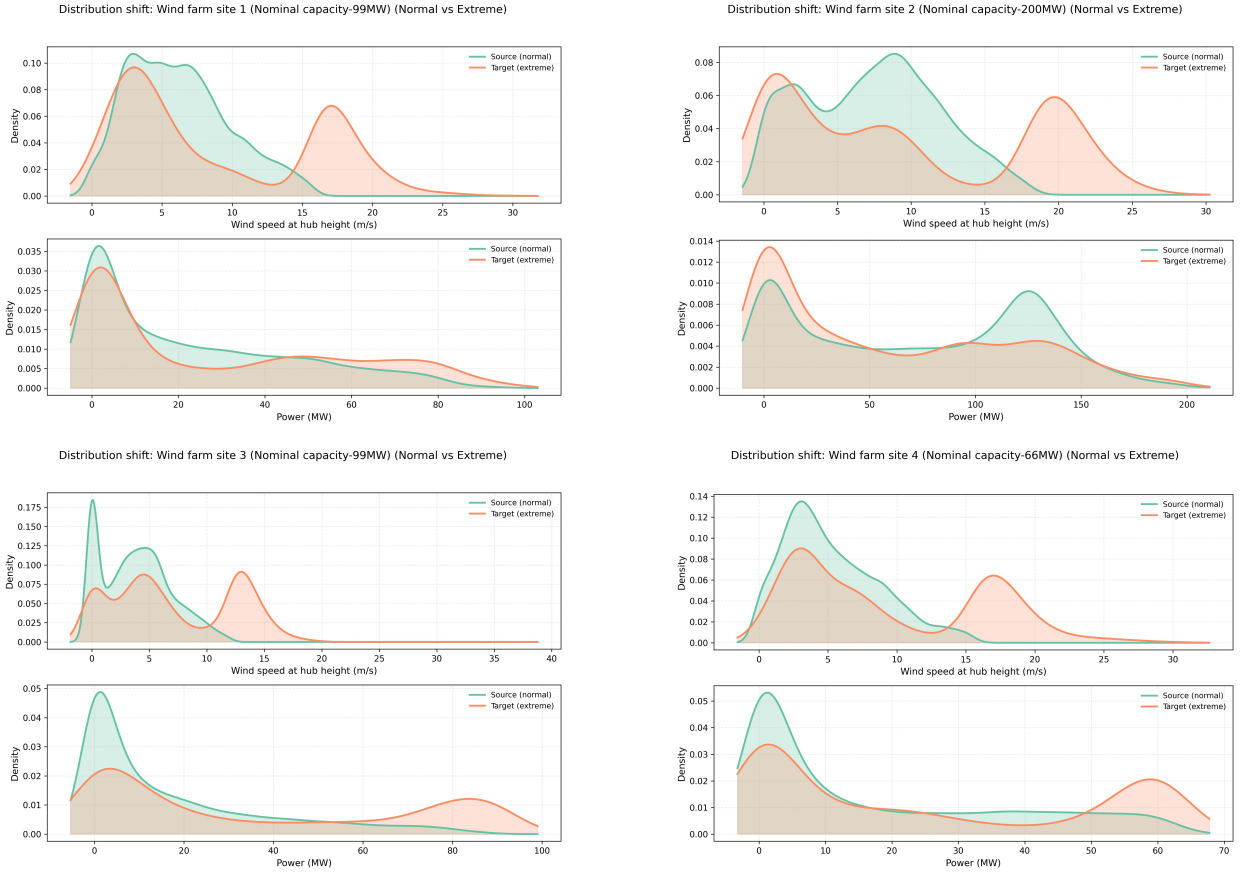
The first experiment identifies extreme-event samples across all six wind farms using the criteria in Section 4.3, and then quantifies source-target distribution shift for each farm. Table 3 reports the domain-shift statistics sorted by the composite Wasserstein distance $W_1(\text{wind}) + W_1(\text{power})$. Preliminary sensitivity checks show that a stricter 99th-percentile wind-speed rule leaves too few target-domain samples for stable model development, whereas the adopted 98th-percentile rule preserves event extremity while maintaining statistical sufficiency.

Fig. 2 visualizes the distribution shift between normal-operating and extreme-event domains for all six wind farms. Each subplot displays the kernel density estimates of wind speed (upper panel) and power output (lower panel) for both domains. The visual comparison confirms the quantitative findings in Table 3: Farm 6 exhibits the most pronounced separation between the two distributions, particularly in the power dimension, where the extreme-event domain shows

a bimodal pattern with substantial mass near zero power (cut-out events) and high power (pre-cut-out high wind conditions).

Several observations emerge from Table 3 and Fig. 2. First, all six farms exhibit non-trivial distribution shifts between the normal-operating and extreme-event domains, with KS statistics ranging from 0.116 to 0.460, confirming that extreme events induce statistically significant distributional changes in both wind speed and power output. Second, the magnitude of the power distribution shift varies substantially across farms: Farm 6 exhibits a Wasserstein distance of 21.057 in the power dimension, approximately 3.5 times that of Farm 1 (5.964). This disparity reflects differences in turbine control strategies, terrain effects, and the local meteorological characteristics of each site.

Farm 6 (nominal capacity 96 MW) ranks first in the composite domain shift metric ($W_1(\text{wind}) + W_1(\text{power}) = 25.117$), indicating the most pronounced distributional discrepancy between its normal-operating and extreme-event regimes. Consequently, Farm 6 is selected as the target wind farm, providing the most stringent test bed for evaluating the robustness of transfer learning methods under extreme-event conditions.



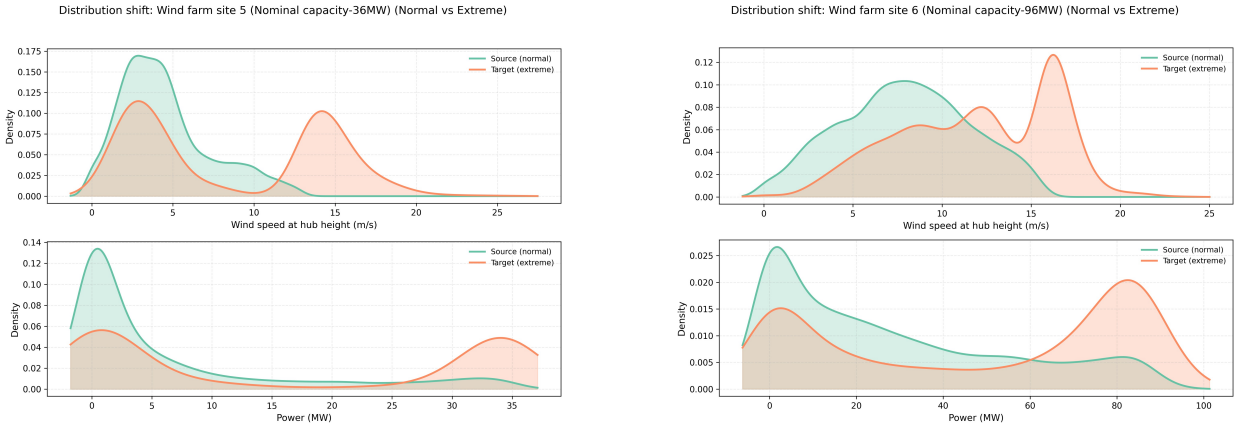


Fig. 2: Distribution shift visualization for all six wind farms. From left to right, top to bottom: Site 1 (99 MW), Site 2 (200 MW), Site 3 (99 MW), Site 4 (66 MW), Site 5 (36 MW), and Site 6 (96 MW). Each subplot shows kernel density estimates of wind speed (upper panel) and power output (lower panel) for the normal-operating (source) and extreme-event (target) domains. Site 6 exhibits the most severe domain shift in the power dimension.

4.6. Experiment 2: Deterministic Point Prediction under Extreme Events

The second experiment evaluates deterministic point prediction on Farm 6 and compares TL-QLDMR with nine baselines under the same data split and preprocessing protocol.

The data partition for Farm 6 is as follows:

- **Source domain** (normal operation): 66,009 samples;
- **Target domain** (extreme-event domain): 4,143 samples.

We compare TL-QLDMR against classical kernel baselines, robust kernel variants, and representative ensemble methods for wind power forecasting:

Classical and robust kernel baselines:

(1) **Standard SVR**: Classical Support Vector Regression with RBF kernel and ϵ-insensitive loss, serving as the foundational kernel-based regression baseline;

(2) **Standard LDMR**: Large-margin Distribution Machine-based Regression with symmetric variance regularization, representing the direct predecessor of TL-QLDMR without transfer learning or asymmetric modifications;

(3) **BLSSVR** (Fu et al., 2023): A bounded-loss SVR-type regression model designed to improve robustness against outliers and heavy-tailed disturbances.

Additional SVM variants:

(4) **HHO-SVR** (Houssein et al., 2025): Support Vector Regression with hyperparameters optimized by the Harris Hawks Optimization metaheuristic;

(5) **FLSVR** (Meena et al., 2025): A functional iterative method for solving the Lagrangian dual of SVR, offering improved convergence properties;

(6) **ARA-SVR** (Duan et al., 2025): SVR with Associated Road Analysis for feature selection, originally proposed for short-term traffic flow prediction.

Ensemble methods:

(7) **KMeans-GBT** (Rizkallah, 2025): A two-stage model combining KMeans clustering with Gradient Boosted Trees to capture heterogeneous input–output relationships;

Table 4
Deterministic point prediction results on the extreme-event test set of Farm 6. Best results are highlighted in bold (similarly hereinafter).

Model	RMSE	MAE	R^2
Kernel-based methods			
Standard LDMR	10.692	6.327	0.9085
BLSSVR	10.273	6.140	0.9155
ARA-SVR	9.246	5.954	0.9316
FLSVR	8.794	6.301	0.9381
Standard SVR	8.564	5.537	0.9413
HHO-SVR	7.603	5.530	0.9537
Ensemble methods			
BRF-WPF	9.719	6.870	0.9244
KMeans-GBT	9.620	6.695	0.9259
RF-WPF	7.967	5.344	0.9492
TL-QLDMR (Ours)	**7.478**	**5.248**	**0.9557**

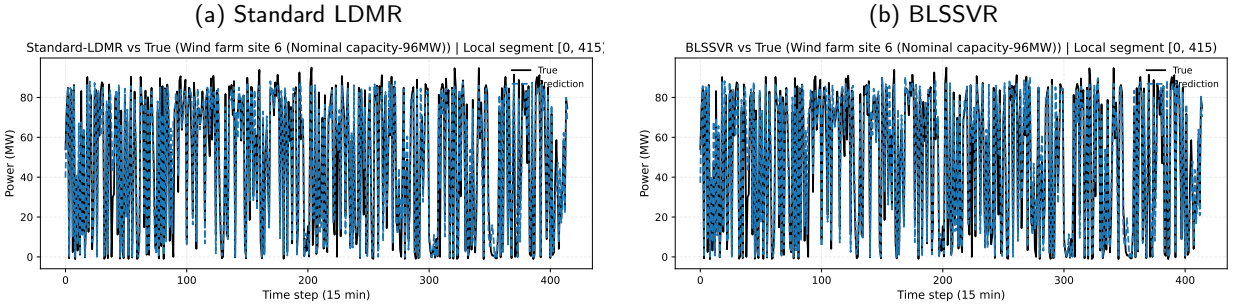


Fig. 3: Prediction comparison over selected temporal segments on the Farm 6 extreme-event test set (Part I).

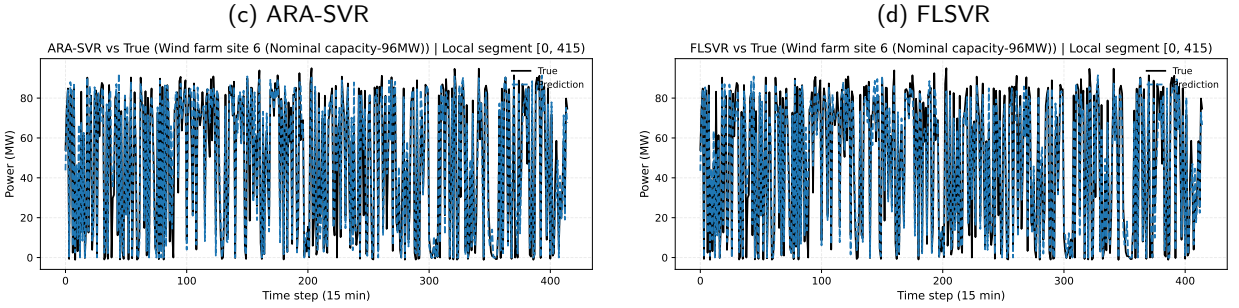


Fig. 4: Prediction comparison over selected temporal segments on the Farm 6 extreme-event test set (Part II).

(8) **RF-WPF** (Mustaffa & Sulaiman, 2025): A Random Forest model tailored for wind power forecasting with optimized tree depth and feature subsampling;

(9) **BRF-WPF** (Chen & Shi, 2025): Broad Random Forest integrating Broad Learning System feature expansion with Random Forest regression for short-term wind power prediction.

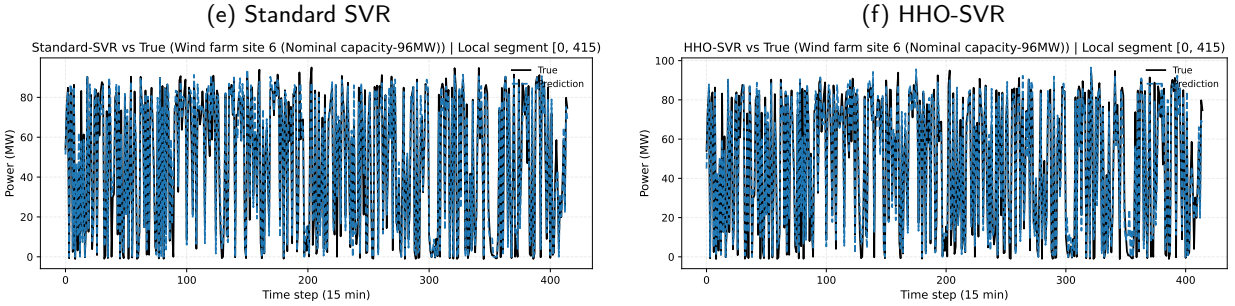


Fig. 5: Prediction comparison over selected temporal segments on the Farm 6 extreme-event test set (Part III).

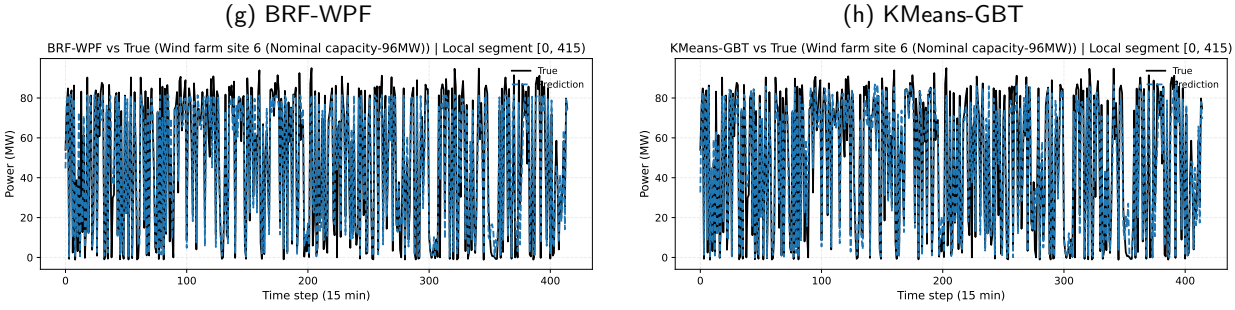


Fig. 6: Prediction comparison over selected temporal segments on the Farm 6 extreme-event test set (Part IV).

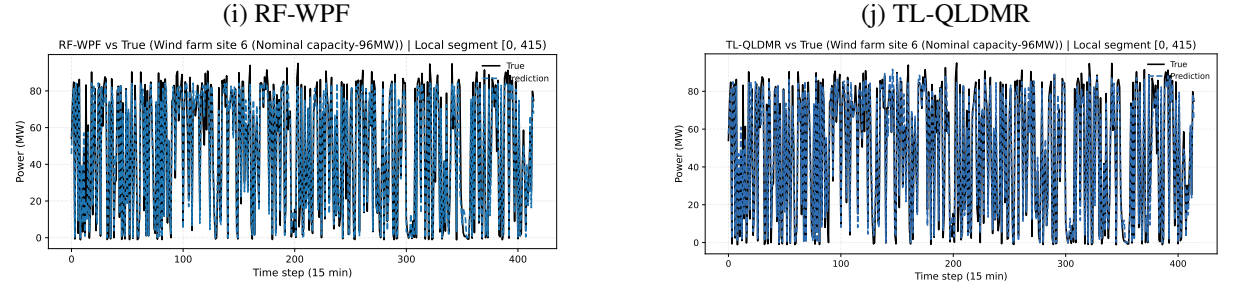


Fig. 7: Prediction comparison over selected temporal segments on the Farm 6 extreme-event test set (Part V).

Table 4 and Figs. 3–7 jointly demonstrate that TL-QLDMR provides the most accurate and stable point forecasts on the Farm 6 extreme-event test set. Several observations are noteworthy:

First, comparing TL-QLDMR against classical and robust kernel baselines reveals the value of the proposed extensions. Standard SVR achieves moderate performance (RMSE = 8.564, $R^2 = 0.9413$), establishing a solid baseline for kernel-based regression. However, Standard LDMR, despite incorporating variance regularization for distributional robustness, exhibits the weakest performance among kernel methods (RMSE = 10.692, $R^2 = 0.9085$). BLSSVR, although designed to improve robustness via bounded loss, improves only marginally over Standard LDMR (RMSE = 10.273, $R^2 = 0.9155$) and remains clearly inferior to Standard SVR and HHO-SVR. This pattern confirms our theoretical analysis in Section 3: sample-wise robustness alone is insufficient in the cross-domain extreme-event setting where (i) training and testing distributions differ substantially, and (ii) the model must adapt to target-domain dynamics. The large performance gaps from Standard LDMR and BLSSVR to TL-QLDMR quantitatively validate the necessity of jointly incorporating transfer learning and asymmetric variance regularization.

Table 5
Model performance (R^2) under varying SNR on the Farm 6 extreme-event test set.

Model	SNR = 30 dB	SNR = 40 dB	SNR = 60 dB
Kernel-based methods			
Standard LDMR	0.9084	0.9080	0.9084
BLSSVR	0.9121	0.9146	0.9154
ARA-SVR	0.9311	0.9316	0.9316
FLSVR	0.9380	0.9381	0.9381
Standard SVR	0.9409	0.9414	0.9413
HHO-SVR	0.9442	0.9438	0.9437
Ensemble methods			
BRF-WPF	0.9177	0.9250	0.9253
KMeans-GBT	0.9209	0.9230	0.9262
RF-WPF	0.9471	0.9485	0.9469
TL-QLDMR (Ours)	**0.9584**	**0.9560**	**0.9603**

Second, TL-QLDMR improves on the strongest non-transfer baselines, including HHO-SVR and RF-WPF. Relative to HHO-SVR, TL-QLDMR raises R^2 from 0.9537 to 0.9557 and reduces RMSE from 7.603 to 7.478. It also improves on RF-WPF, the strongest ensemble comparator, reducing RMSE from 7.967 to 7.478. Although the numerical margin over HHO-SVR is moderate, it is obtained under the most severe domain-shift setting with only 4,143 extreme-event samples. The segment-level evidence in Figs. 3–7 is consistent with the aggregate metrics: TL-QLDMR tracks abrupt ramps and low-power transitions more closely, while weaker kernel baselines such as Standard LDMR and BLSSVR show larger local deviations around rapid fluctuations.

Third, among baselines, HHO-SVR remains the most competitive kernel method, RF-WPF remains the strongest non-kernel alternative, and BLSSVR does not translate its bounded-loss robustness into competitive cross-domain accuracy under the present small-sample extreme-event regime. The observed gain of TL-QLDMR is consistent with the combined effect of source-domain transfer, asymmetric target weighting, and cyclic wind-direction encoding. Because many extreme-event periods contain near-zero power outputs, MAPE is numerically unstable and therefore RMSE and R^2 are treated as the principal deterministic indicators.

To assess the robustness of the models against measurement noise, which is prevalent in SCADA systems during extreme events, we evaluate the point prediction performance by injecting Gaussian noise into both the training and testing input features at Signal-to-Noise Ratios SNR $\in \{60, 40, 30\}$ dB. The R^2 results under different noise levels are summarized in Table 5.

The results confirm that TL-QLDMR consistently maintains the highest R^2 across all noise levels, achieving 0.9584, 0.9560, and 0.9603 at SNR = 30, 40, and 60 dB, respectively. The strongest baseline under noisy inputs is HHO-SVR, but it remains below TL-QLDMR at all three settings ($R^2 = 0.9442, 0.9438$, and 0.9437). BLSSVR remains slightly better than Standard LDMR under noise ($R^2 = 0.9121$ vs. 0.9084 at SNR = 30 dB), but still trails the stronger kernel baselines by a clear margin, again indicating that bounded-loss robustness alone does not resolve the cross-domain mismatch. Standard SVR demonstrates moderate robustness ($R^2 = 0.9409$ at SNR = 30 dB), outperforming both Standard LDMR and BLSSVR but remaining substantially below TL-QLDMR. Among ensemble methods, BRF-WPF exhibits the steepest degradation, indicating higher sensitivity to measurement perturbations. The transfer learning framework, reinforced by asymmetric variance regularization, effectively mitigates overfitting to noisy target samples and demonstrates the strongest noise resilience among all comparative models.

4.7. Experiment 3: Probabilistic Interval Prediction under Extreme Events

The third experiment extends evaluation from point prediction to probabilistic interval forecasting. The objective is to verify whether TL-QLDMR can achieve reliable 95% prediction intervals while keeping interval width sufficiently sharp under extreme-event volatility.

The interval prediction baselines are organized into two categories:

Kernel-based quantile and robust learning variants:

(1) **Standard SVQR**: Classical Support Vector Quantile Regression with pinball loss, serving as the foundational kernel-based interval prediction baseline;

(2) **TSVQR** (Ye et al., 2023): Twin Support Vector Quantile Regression, which constructs separate hyperplanes for upper and lower quantiles;

(3) **UQSVM-SSVQR** (Hazarika et al., 2023): A robust SVQR variant with truncated pinball loss, used as a strong SVM-based interval baseline;

(4) **NFS-SVQR** (Ye et al., 2025b): Nonlinear Feature Selection combined with SVQR, published in *Neural Networks* (2025);

(5) **NuSVR-CI** (Romano et al., 2019): ν-Support Vector Regression with conformalized quantile regression (CQR)-calibrated prediction intervals, included as a classical SVM baseline;

(6) **RLMKL** (Jiang et al., 2021): Robust low-rank multiple kernel learning with compound regularization; its quantile-loss instantiation is used here as a strong robust kernel interval baseline.

Recent wind power interval prediction models:

(7) **HybridDL-Interval** (Mao et al., 2025): A hybrid deep learning framework for day-ahead wind power interval forecasting, combining temporal attention and quantile loss;

(8) **AMQRNN** (Ye et al., 2025a): A recurrent attention encoder-decoder network for multi-step interval wind power prediction, implemented using an attention-GRU network;

(9) **QGBR** (Hu & Liu, 2025): Quantile Gradient Boosted Regression, representing the improved quantile ensemble learning framework for wind power prediction;

(10) **CQR-MLP** (Hu et al., 2025): MLP quantile ensemble with dynamic conformalized quantile regression, acting as a simplified baseline for neural ensemble search methods.

Table 6 presents the interval prediction results on the extreme-event test set of Farm 6 at the 95% confidence level ($\alpha = 0.05$).

The interval prediction results indicate a clear reliability-sharpness trade-off under extreme events, and TL-QLDMR offers the most favorable balance. TL-QLDMR reaches the nominal 95% coverage target (PICP = 0.959) while obtaining the best composite quality (CWC = 0.268) and the best Winkler score (34.506), demonstrating that transfer learning with validation-based width scaling can provide both reliable and sharp intervals.

Comparing against the kernel-based interval baselines highlights two distinct patterns. Standard SVQR achieves the same coverage level as TL-QLDMR (PICP = 0.959) but produces substantially wider intervals (PINAW = 0.357 vs. 0.268), resulting in inferior CWC (0.357 vs. 0.268) and Winkler scores (42.067 vs. 34.506). RLMKL, as a robust multiple-kernel comparator, attains reliable coverage (PICP = 0.957) with noticeably narrower intervals than Standard SVQR (PINAW = 0.331 vs. 0.357) and a better Winkler score (39.275 vs. 42.067), confirming the value of robust kernel combination. Nevertheless, TL-QLDMR still provides a materially sharper and better-calibrated solution, reducing CWC from 0.331 to 0.268 while further improving the Winkler score to 34.506.

Table 6
Interval prediction results on the extreme-event test set of Farm 6 at 95% confidence level ($\alpha = 0.05$).

Model	PICP	MPIW	PINAW	CWC	Winkler
NuSVR-CI	0.843	**19.970**	**0.208**	0.285	54.603
HybridDL-Interval	0.954	31.645	0.329	0.329	60.224
RLMKL	0.957	31.804	0.331	0.331	39.275
TSVQR	0.889	24.592	0.256	0.342	48.178
Standard SVQR	**0.959**	34.342	0.357	0.357	42.067
QGBR	0.949	31.715	0.330	0.429	37.643
CQR-MLP	0.942	31.750	0.330	0.431	46.772
AMQRNN	0.940	32.432	0.337	0.441	66.874
NFS-SVQR	0.913	68.359	0.711	0.941	76.997
UQSVM-SSVQR	0.928	81.710	0.850	1.117	85.162
TL-QLDMR (Ours)	**0.959**	25.791	0.268	**0.268**	**34.506**

By contrast, NuSVR-CI and TSVQR produce narrow intervals but severe under-coverage (PICP = 0.843 and 0.889, respectively), whereas HybridDL-Interval, QGBR, CQR-MLP, and AMQRNN can approach acceptable coverage only with wider intervals. RLMKL therefore emerges as a stronger robust-kernel baseline than the conventional quantile regressors, but it still remains dominated by TL-QLDMR on the joint reliability-sharpness criteria. This comparison suggests that explicit asymmetric regularization combined with cross-domain adaptation is more effective for heavily skewed extreme-event samples than either conventional quantile regression, robust multiple-kernel learning alone, or deep learning approaches.

To assess interval robustness under sensor perturbation, Gaussian noise is injected into test inputs at SNR $\in \{60, 40, 30\}$ dB. Table 7 reports the full set of interval metrics under this noisy-test protocol. Across all noise levels, TL-QLDMR maintains the best CWC while keeping PICP above 0.95. The CWC values remain tightly concentrated around 0.278, indicating stable interval quality under moderate measurement perturbation. RLMKL remains one of the more stable robust-kernel baselines, with PINAW staying near 0.313 across all noise levels, but its PICP remains slightly below the nominal level (0.9494), leading to a larger CWC penalty than TL-QLDMR. In contrast, NuSVR-CI yields narrow intervals (small MPIW/PINAW) but exhibits severe under-coverage, and several deep/ensemble baselines require substantially wider intervals to approach reliable coverage. These results indicate that TL-QLDMR preserves a stronger reliability-sharpness balance under measurement perturbations.

4.8. Experiment 4: Ablation Study

To validate the contribution of the three core components (transfer learning, LDMR variance regularization, and MMD feature alignment), this ablation experiment evaluates structurally reduced variants on both point and interval tasks. Additionally, to verify the necessity of asymmetric variance regularization for quantile regression, we include LDMR-Quantile (SymVar), which replaces the asymmetric variance term with standard symmetric variance minimization while retaining the transfer learning framework.

The full TL-QLDMR framework is compared against four architectural reductions:

(1) **LDMR-Quantile (SymVar)**: Replaces asymmetric variance regularization with symmetric variance minimization, testing whether the asymmetry is necessary for quantile-specific calibration.

(2) **w/o Transfer (Target-only QLDMR)**: Trained exclusively on extreme-event target samples without source-domain integration or MMD alignment.

(3) **w/o LDMR (Variance Penalty)**: The asymmetric variance regularizer is removed, deteriorating the framework into a generalized Transfer SVR/SVQR variant that lacks distributional robustness.

Table 7
Interval prediction results under varying SNR on the extreme-event test set of Farm 6 at 95% confidence level ($\alpha = 0.05$).

Model	SNR (dB)	PICP	MPIW	PINAW	CWC	Winkler
	30	0.8386	**19.970**	**0.2077**	0.2856	54.922
NuSVR-CI	40	0.8361	**19.970**	**0.2077**	0.2860	54.623
	60	0.8434	**19.970**	**0.2077**	0.2849	54.588
	30	0.9422	32.304	0.3360	0.4384	53.462
HybridDL-Interval	40	0.9422	32.307	0.3361	0.4385	53.420
	60	0.9422	32.306	0.3361	0.4385	53.408
	30	0.9494	30.161	0.3138	0.4080	39.063
RLMKL	40	0.9494	30.101	0.3131	0.4072	39.235
	60	0.9494	30.095	0.3131	0.4071	39.200
	30	0.8892	24.570	0.2556	0.3422	48.306
TSVQR	40	0.8892	24.588	0.2558	0.3424	48.140
	60	0.8892	24.592	0.2558	0.3425	48.169
	30	0.9446	29.704	0.3090	0.4027	41.541
Standard SVQR	40	0.9422	29.725	0.3092	0.4034	41.429
	60	0.9446	29.728	0.3092	0.4030	41.420
	30	0.9518	32.067	0.3336	0.3336	38.441
QGBR	40	0.9518	31.721	0.3300	0.3300	37.424
	60	0.9470	31.706	0.3298	0.4294	37.750
	30	0.9325	32.548	0.3386	0.4438	48.828
CQR-MLP	40	0.9325	32.550	0.3386	0.4438	48.780
	60	0.9325	32.554	0.3386	0.4438	48.743
	30	0.9518	33.517	0.3487	0.3487	68.546
AMQRNN	40	0.9518	33.513	0.3486	0.3486	68.517
	60	0.9518	33.512	0.3486	0.3486	68.515
	30	0.9060	68.369	0.7112	0.9442	77.305
NFS-SVQR	40	0.9133	68.376	0.7113	0.9409	76.878
	60	0.9133	68.359	0.7111	0.9407	76.983
	30	0.9229	81.706	0.8499	1.1191	85.364
UQSVM-SSVQR	40	0.9277	81.718	0.8501	1.1167	85.178
	60	0.9277	81.710	0.8500	1.1166	85.162
	30	**0.9542**	26.709	0.2778	**0.2778**	**36.307**
TL-QLDMR (Ours)	40	**0.9566**	26.713	0.2779	**0.2779**	**36.034**
	60	**0.9566**	26.715	0.2779	**0.2779**	**36.083**

(4) **w/o MMD (Feature Alignment)**: The algorithm employs cross-domain training data but disables Maximum Mean Discrepancy alignment, relying purely on weighted objective loss.

Table 8 reports the ablation results on Farm 6. The full TL-QLDMR consistently attains the best deterministic and probabilistic performance, indicating that all three components contribute non-trivially to final accuracy and calibration quality.

Table 8
Ablation study evaluations for point and interval prediction on the Farm 6 test set.

	Point Prediction		Interval Prediction (95% Level)			
Model Variants	RMSE	R^2	PICP	MPIW	PINAW	CWC
w/o LDMR (Variance)	8.205	0.9461	0.949	25.925	0.2697	0.3507
w/o Transfer	7.930	0.9497	0.952	26.891	0.2797	0.2797
LDMR-Quantile (SymVar)	7.868	0.9472	0.952	29.114	0.3029	0.3029
w/o MMD	7.745	0.9520	0.947	26.049	0.2710	0.3528
TL-QLDMR (Full)	**7.476**	**0.9557**	**0.959**	**25.791**	**0.2683**	**0.2683**

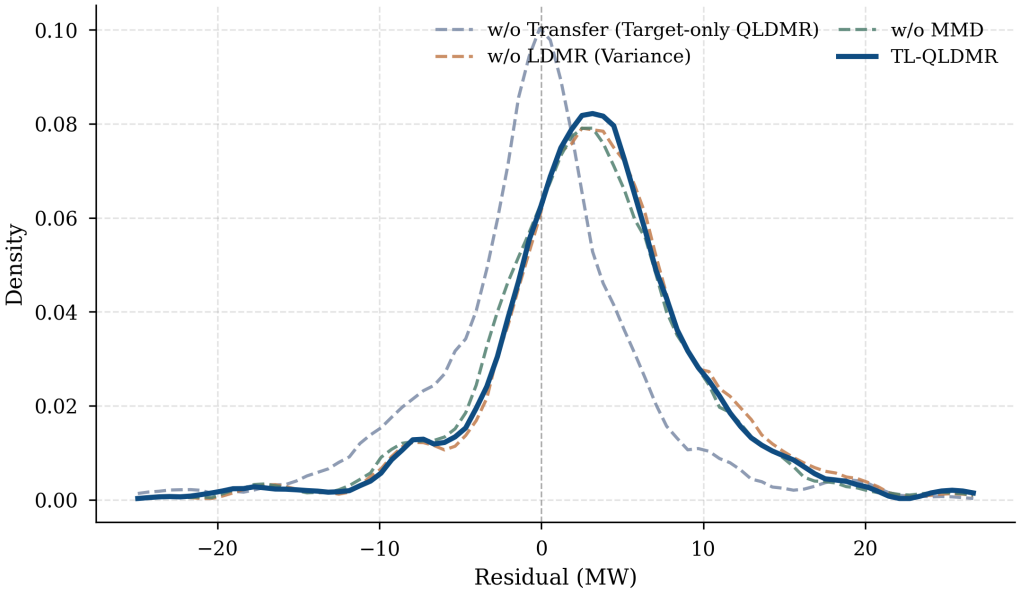


Fig. 8: Residual density comparison between TL-QLDMR and ablated variants on the Farm 6 test set.

The ablation outcomes confirm that each component contributes materially to robustness. Removing transfer learning reduces both point accuracy (R^2: 0.9557 → 0.9497) and interval quality (CWC: 0.2683 → 0.2797), indicating that target-only training is insufficient under strong domain shift. Removing LDMR regularization or MMD alignment causes clear interval-quality degradation (CWC rises to 0.3507 and 0.3528), even when point-metric changes are smaller, showing that residual-shape control and feature alignment are central to reliable uncertainty estimation.

Critically, the LDMR-Quantile (SymVar) variant, which replaces asymmetric variance with symmetric variance minimization, exhibits degraded performance on both point prediction (RMSE: 7.868 vs. 7.476) and interval prediction (CWC: 0.3029 vs. 0.2683, PINAW: 0.3029 vs. 0.2683). This degradation empirically validates our theoretical analysis in Section 3: symmetric variance minimization, while effective for mean regression, conflicts with the asymmetric residual distribution required for accurate quantile estimation. The symmetric regularizer forces errors toward a centered, Gaussian-like distribution, thereby degrading interval calibration at non-median quantiles.

The residual-density comparison in Fig. 8 supports this conclusion by showing tighter central concentration and weaker heavy tails for the full TL-QLDMR model; some ablated variants exhibit sharper central peaks but noticeably thicker tails, which corresponds to a higher probability of large-error events.

Table 9
Training time (seconds) of four solvers over expanding training scales on Farm 6. "—" denotes runs not extended further because of prohibitive computational cost.

N	CVXOPT	Torch-GD	Batch SGD	Nyström-SGD (Ours)
100	0.509	2.155	2.746	**0.035**
500	1.052	2.618	4.637	**1.035**
1,000	7.655	2.979	5.699	**1.038**
3,000	—	20.395	6.418	**2.468**
10,000	—	—	10.401	**2.703**
30,000	—	—	25.338	**3.483**
69,737	—	—	—	**5.021**

4.9. Experiment 5: Scalable Optimization Validation

Experiment 5 evaluates whether the proposed Nyström-SGD solver can maintain computational efficiency as the training scale increases. The proposed solver is compared against three representative optimization baselines: (i) exact dual quadratic programming solved by CVXOPT, (ii) gradient-based optimization of the QP objective (Torch-GD), and (iii) mini-batch SGD with kernel-slice approximation.

To stress-test scalability, we consider seven training scales, increasing the combined training size from 100 to the full $N = 69{,}737$. The evaluated sizes are $N = 100, 500, 1{,}000, 3{,}000, 10{,}000, 30{,}000$, and 69,737. For each configuration, wall-clock training time (seconds) is recorded. Table 9 summarizes the results.

The full-QP routes lose practicality quickly as the scale increases. CVXOPT grows from 0.509 seconds at $N = 100$ to 7.655 seconds at $N = 1{,}000$ and is not extended further thereafter. Torch-GD remains workable slightly longer, but already requires 20.395 seconds at $N = 3{,}000$, indicating that direct optimization of the kernel QP becomes inefficient well before the full training regime is reached.

Batch SGD avoids exact QP solving but remains substantially slower than the proposed Nyström-SGD solver at medium and large scales; for example, at $N = 30{,}000$ it requires 25.338 seconds, compared with 3.483 seconds for the Nyström-SGD route. By contrast, the proposed solver is the only optimization route that completes the entire uncapped training set, reaching $N = 69{,}737$ in 5.021 seconds. This trend is consistent with the theoretical reduction from $O(N^3)$ to $O(T \cdot N \cdot m)$: runtime growth remains comparatively mild even after the target-domain sample count saturates and the additional data come almost entirely from the source domain.

4.10. Discussion

The extended experimental evaluations yield the following conclusive insights:

Classical and robust kernel baselines establish the necessity of proposed extensions. Experiment 2 demonstrates that Standard SVR achieves moderate performance (RMSE = 8.564), while Standard LDMR and BLSSVR remain substantially weaker on the cross-domain extreme-event task. This degradation confirms that symmetric variance minimization or bounded-loss robustness alone is insufficient when target-domain adaptation is absent. In Experiment 3, RLMKL provides a stronger robust-kernel interval baseline than conventional SVQR variants, but TL-QLDMR still achieves markedly better joint reliability-sharpness quality, showing that asymmetric variance regularization and supervised transfer adaptation remain necessary.

Transfer learning improves deterministic forecasting under severe shift. Experiment 2 shows consistent gains in RMSE and R^2 over both classical baselines and recent variants, and curve-level evidence confirms better tracking during abrupt peak-valley transitions.

Interval forecasting achieves a better reliability-sharpness balance. In Experiment 3, TL-QLDMR achieves the highest coverage reliability among the compared methods while also obtaining the best composite penalty (CWC), and this advantage persists under injected measurement noise.

Ablation confirms the necessity of each component. Removing transfer learning reduces both point and interval performance under severe domain shift; removing asymmetric LDMR variance regularization degrades distributional robustness and interval calibration; removing MMD alignment weakens feature-space adaptation. Critically, the LDMR-Quantile (SymVar) variant confirms that symmetric variance minimization conflicts with quantile-specific asymmetry requirements, validating our theoretical analysis. Residual-density plots provide complementary evidence for this degradation.

The scalable optimizer materially reduces computational burden. Experiment 5 shows that the Nyström-SGD solver is the only optimization route that reaches the full uncapped training scale ($N = 69{,}737$), completing training in 5.021 seconds, whereas CVXOPT, Torch-GD, and batch SGD stall much earlier due to rapidly increasing computational cost.

Limitations and future work. The present evaluation is based on onshore wind farms and fixed feature engineering pipelines; extension to offshore assets, dynamic re-calibration strategies, and hybrid deep-kernel transfer formulations remains a relevant direction.

5. Conclusion

This paper presents TL-QLDMR for wind power forecasting under extreme events, where target-domain samples are limited and distribution shift is pronounced. The method combines asymmetric variance regularization, MMD-based feature alignment, and weighted pinball loss within a unified transfer-learning quantile framework, with theoretical guarantees on positive definiteness and convergence.

Experimental results show that TL-QLDMR delivers consistent gains in both deterministic and interval forecasting on the most challenging target farm. For point prediction, it achieves the best RMSE (7.478) and R^2 (0.9557). For 95% interval prediction, it attains PICP $= 0.959$ together with the best CWC (0.268), indicating a favorable reliability-sharpness trade-off. Under injected noise, TL-QLDMR remains the strongest model across reported point and interval metrics. In addition, the Nyström-SGD solver provides practical scalability, completing the full uncapped training set ($N = 69{,}737$) in 5.021 seconds, whereas CVXOPT, Torch-GD, and batch SGD do not remain computationally viable at the same scale.

Future work will focus on extending the framework to multi-site spatiotemporal coupling, adaptive online recalibration under non-stationary weather, and hybrid kernel-deep transfer formulations.

CRediT authorship contribution statement

Junlin Chen: Methodology, Software, Writing - original draft. **Bo Wang:** Writing – review & editing, Funding acquisition. **Zhihang Yu:** Formal analysis, Software. **Renju Lin:** Software. **Pei-chun Lin:** Validation, Supervision. **Libo Zhang:** Funding acquisition. **Junzo Watada:** Validation, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China under Grant 72471109 and Grant 62106205.

References

Ben-David, S., Blitzer, J., Crammer, K., Kulesza, A., Pereira, F., & Vaughan, J. W. (2010). A theory of learning from different domains. *Machine Learning*, *79*, 151–175. doi:10.1007/s10994-009-5152-4.

Bottou, L., Curtis, F. E., & Nocedal, J. (2018). Optimization methods for large-scale machine learning. *SIAM Review*, *60*, 223–311. doi:10.1137/16M1080173.

Chen, Y., & Shi, J. (2025). Broad random forest: A lightweight prediction model for short-term wind power by fusing broad learning and random forest. *Sustainability*, *17*, 4894. doi:10.3390/su17114894.

Chen, Y., & Xu, J. (2022). Solar and wind power data from the chinese state grid renewable energy generation forecasting competition. *Scientific Data*, *9*, 684. doi:10.1038/s41597-022-01696-6.

Cheng, F., Zhang, J., & Wen, C. (2016). Cost-sensitive large margin distribution machine for classification of imbalanced data. *Pattern Recognition Letters*, *80*, 107–112. https://doi.org/10.1016/j.patrec.2016.06.009.

Dong, X., Sun, Y., Dong, L., Li, J., Li, Y., & Di, L. (2023). Transferable wind power probabilistic forecasting based on multi-domain adversarial networks. *Energy*, *285*, 129496. doi:10.1016/j.energy.2023.129496.

Duan, Y. et al. (2025). Research on support vector regression short-time traffic flow prediction model for secondary roads based on associated road analysis. *Applied Sciences*, *15*, 1779. doi:10.3390/app15041779.

Fu, S., Tian, Y., & Tang, L. (2023). Robust regression under the general framework of bounded loss functions. *European Journal of Operational Research*, *310*, 1325–1339. doi:10.1016/j.ejor.2023.04.025.

Guo, T., Tan, X., Yang, L., Liang, Z., Zhang, B., & Zhang, L. (2022). Domain adaptive subspace transfer model for sensor drift compensation in biologically inspired electronic nose. *Expert Systems with Applications*, *208*, 118237. https://doi.org/10.1016/j.eswa.2022.118237.

Gupta, U., & Gupta, D. (2021). Least squares large margin distribution machine for regression. *Applied Intelligence*, *51*, 7058–7093. doi:10.1007/s10489-020-02166-5.

Hazarika, B. B., & Gupta, D. (2023). Mode decomposition based large margin distribution machines for sediment load prediction. *Expert Systems with Applications*, *232*, 120844. doi:10.1016/j.eswa.2023.120844.

Hazarika, B. B., Gupta, D., & Borah, P. (2023). Robust support vector quantile regression with truncated pinball loss (rsvqr). *Computational and Applied Mathematics*, *42*. doi:10.1007/s40314-023-02402-x.

Houssein, E. H. et al. (2025). A hybrid harris hawks optimization with support vector regression for air quality forecasting. *Scientific Reports*, *15*. doi:10.1038/s41598-025-86275-6.

Hu, J., Deng, Y., & Che, J. (2025). A novel wind power interval prediction method based on neural ensemble search and dynamic conformalized quantile regression. *Applied Soft Computing*, *180*, 113476. doi:10.1016/j.asoc.2025.113476.

Hu, L., & Liu, Y. (2025). A day-ahead wind power conformal interval forecasting framework based on numerical weather prediction data and improved quantile ensemble learning. *Journal of Renewable and Sustainable Energy*, *17*, 046104. doi:10.1063/5.0233770.

Jiang, H., Tao, C., Dong, Y., & Xiong, R. (2021). Robust low-rank multiple kernel learning with compound regularization. *European Journal of Operational Research*, *295*, 634–647. doi:10.1016/j.ejor.2020.12.024.

Jonkers, J., Avendano, D. N., Van Wallendael, G., & Van Hoecke, S. (2024). A novel day-ahead regional and probabilistic wind power forecasting framework using deep cnns and conformalized regression forests. *Applied Energy*, *361*, 122900. doi:10.1016/j.apenergy.2024.122900.

Kumar, S., Mohri, M., & Talwalkar, A. (2012). Sampling methods for the nyström method. *Journal of Machine Learning Research*, *13*, 981–1006.

Li, L.-L., Zhao, X., Tseng, M.-L., & Tan, R. R. (2021). Improving short-term wind power prediction using hybrid improved cuckoo search arithmetic – support vector regression machine. *Journal of Cleaner Production*, *279*, 123739. doi:10.1016/j.jclepro.2020.123739.

Liang, Z., Chai, R., Sun, Y., Jiang, Y., & Sun, D. (2025). Cold wave recognition and wind power forecasting technology considering sample scarcity and meteorological periodicity characteristics. *Applied Sciences*, *15*, 4312. doi:10.3390/app15084312.

Liu, G., Liu, J., Bai, Y., Wang, C., Wang, H., Zhao, H., Liang, G., Zhao, J., & Qiu, J. (2023). Eweld: A large-scale industrial and commercial load dataset in extreme weather events. *Scientific Data*, *10*, 615. doi:10.1038/s41597-023-02503-6.

Liu, Y., Wang, J., & Liu, L. (2024). Physics-informed reinforcement learning for probabilistic wind power forecasting under extreme events. *Applied Energy*, *376*, 124068. doi:10.1016/j.apenergy.2024.124068.

Luo, J., Hong, T., Gao, Z., & Fang, S.-C. (2024). Fuzzy support vector regressions for short-term load forecasting. *Fuzzy Optimization and Decision Making*, *23*, 363–385. doi:10.1007/s10700-024-09422-0.

Mao, Y., Song, L., & Huang, H. (2025). A novel hybrid deep learning model for day-ahead wind power interval forecasting. *Sustainability*, *17*, 3239. doi:10.3390/su17073239.

Meena, Y. et al. (2025). FLSVR: Solving lagrangian support vector regression using functional iterative method. *Neural Processing Letters*, *57*, 71. doi:10.1007/s11063-025-11780-8.

Mustaffa, Z., & Sulaiman, M. H. (2025). Random forest based wind power prediction method for sustainable energy system. *Cleaner Energy Systems*, *13*, 100210. doi:10.1016/j.cles.2025.100210.

Pan, S. J., & Yang, Q. (2010). A survey on transfer learning. *IEEE Transactions on Knowledge and Data Engineering*, *22*, 1345–1359. doi:10.1109/TKDE.2009.191.

Rabanal, A., Ulazia, A., Ibarra-Berastegi, G., Saenz, J., & Elosegui, U. (2019). Midas: A benchmarking multi-criteria method for the identification of defective anemometers in wind farms. *Energies*, *12*, 28. doi:10.3390/en12010028.

Rizkallah, S. (2025). Enhancing the performance of gradient boosting trees on regression problems. *Journal of Big Data*, *12*, 35. doi:10.1186/s40537-025-01071-3.

Romano, Y., Patterson, E., & Candès, E. J. (2019). Conformalized quantile regression. In *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)* (pp. 3538–3548). URL: https://papers.nips.cc/paper/8613-conformalized-quantile-regression.

Schölkopf, B., Herbrich, R., & Smola, A. J. (2001). A general representer theorem. In *Proceedings of the 14th Annual Conference on Computational Learning Theory* (pp. 416–426). Springer. doi:10.1007/3-540-44581-1_27.

Se, H., Song, K., Liu, H., Zhang, W., Wang, X., & Liu, J. (2023). A dual drift compensation framework based on subspace learning and cross-domain adaptive extreme learning machine for gas sensors. *Knowledge-Based Systems*, *259*, 110024. https://doi.org/10.1016/j.knosys.2022.110024.

Sideratos, G., & Hatziargyriou, N. D. (2012). Wind power forecasting focused on extreme power system events. *IEEE Transactions on Sustainable Energy*, *3*, 445–454. doi:10.1109/TSTE.2012.2189442.

Tang, M., Ding, S. X., Yang, C., Cheng, F., Shardt, Y. A., Long, W., & Liu, D. (2019). Cost-sensitive large margin distribution machine for fault detection of wind turbines. *Cluster Computing*, *22*, 7525–7537. doi:10.1007/s10586-018-1854-3.

Tao, J., lai Chung, F., & Wang, S. (2012). On minimum distribution discrepancy support vector machine for domain adaptation. *Pattern Recognition*, *45*, 3962–3984. https://doi.org/10.1016/j.patcog.2012.04.014.

Taylor, J. W. (2007). Forecasting daily supermarket sales using exponentially weighted quantile regression. *European Journal of Operational Research*, *178*, 154–167. doi:10.1016/j.ejor.2006.02.006.

Trafalis, T. B., & Gilbert, R. C. (2006). Robust classification and regression using support vector machines. *European Journal of Operational Research*, *173*, 893–909. doi:10.1016/j.ejor.2005.07.024.

Wang, S., Sun, Y., Zhang, W., & Srinivasan, D. (2025). Optimization of deterministic and probabilistic forecasting for wind power based on ensemble learning. *Energy*, *319*, 134884. doi:10.1016/j.energy.2025.134884.

Wang, S., Zhang, W., Sun, Y., Trivedi, A., Chung, C. Y., & Srinivasan, D. (2024). Wind power forecasting in the presence of data scarcity: A very short-term conditional probabilistic modeling framework. *Energy*, *291*, 130305. doi:10.1016/j.energy.2024.130305.

Wang, Y., Zou, R., Liu, F., Zhang, L., & Liu, Q. (2021). A review of wind speed and wind power forecasting with deep neural networks. *Applied Energy*, *304*, 117766. doi:10.1016/j.apenergy.2021.117766.

Yan, J., Chen, F., Liu, T., Zhang, Y., Peng, X., Yi, D., & Duan, S. (2022). Subspace alignment based on an extreme learning machine for electronic nose drift compensation. *Knowledge-Based Systems*, *235*, 107664. https://doi.org/10.1016/j.knosys.2021.107664.

Yan, J., Li, X., Wang, H., Si, F., Qiao, W., & Liu, Y. (2025). The economic value of wind power forecasting: A data-driven method and its application in various scenarios. *Energy*, *340*, 139151. doi:10.1016/j.energy.2025.139151.

Yang, W., Zhang, Z., & Chen, X. (2024). Vibration-based looseness identification of bolted structures via quasi-analytic wavelet packet and optimized large margin distribution machine. *Structural Health Monitoring*, *23*, 856–875. doi:10.1177/14759217231159948.

Ye, X., Liu, C., Xiong, X., & Qi, Y. (2025a). Recurrent attention encoder-decoder network for multi-step interval wind power prediction. *Energy*, *315*, 134317. doi:10.1016/j.energy.2024.134317.

Ye, Y.-F., Wang, J., & Chen, W.-J. (2025b). Nonlinear feature selection for support vector quantile regression. *Neural Networks*, *185*, 107136. doi:10.1016/j.neunet.2025.107136.

Ye, Y.-F. et al. (2023). Twin support vector quantile regression. *Expert Systems with Applications*, *221*, 121239. doi:10.1016/j.eswa.2023.121239.

Yin, H., Ou, Z., Fu, J., Cai, Y., Chen, S., & Meng, A. (2021). A novel transfer learning approach for wind power prediction based on a serio-parallel deep learning architecture. *Energy*, *234*, 121271. doi:10.1016/j.energy.2021.121271.

Yuan, R., Wang, B., Sun, Y., Song, X., & Watada, J. (2023). Conditional style-based generative adversarial networks for renewable scenario generation. *IEEE Transactions on Power Systems*, *38*, 1281–1296. doi:10.1109/TPWRS.2022.3170992.

Yuan, X., Shi, J., & Xu, J. (2022). Wind power prediction method: Support vector regression optimized by improved jellyfish search algorithm. *Energies*, *15*, 6404. doi:10.3390/en15176404.

Zhai, S., Han, M., Li, Z., Yang, S., Duan, S., & Yan, J. (2023). M2fl-ccc: Multibranch multilayer feature leaning and comprehensive classification criterion for gas sensor drift compensation. *IEEE Transactions on Instrumentation and Measurement*, *72*, 1–12. https://doi.org/10.1109/TIM.2023.3296820.

Zhang, K., & Kwok, J. T. (2010). Clustered nyström method for large scale manifold learning and dimension reduction. *IEEE Transactions on Neural Networks*, *21*, 1576–1587. doi:10.1109/TNN.2010.2064786.

Zhang, T., & Zhou, Z.-H. (2014). Large margin distribution machine. In *Proceedings of the 20th ACM SIGKDD international conference on Knowledge discovery and data mining* (pp. 313–322). doi:10.1145/2623330.2623710.

Zhang, T., & Zhou, Z.-H. (2020). Optimal margin distribution machine. *IEEE Transactions on Knowledge and Data Engineering*, *32*, 1143–1156. doi:10.1109/TKDE.2019.2897662.